



Data Handling: Import, Cleaning and Visualisation

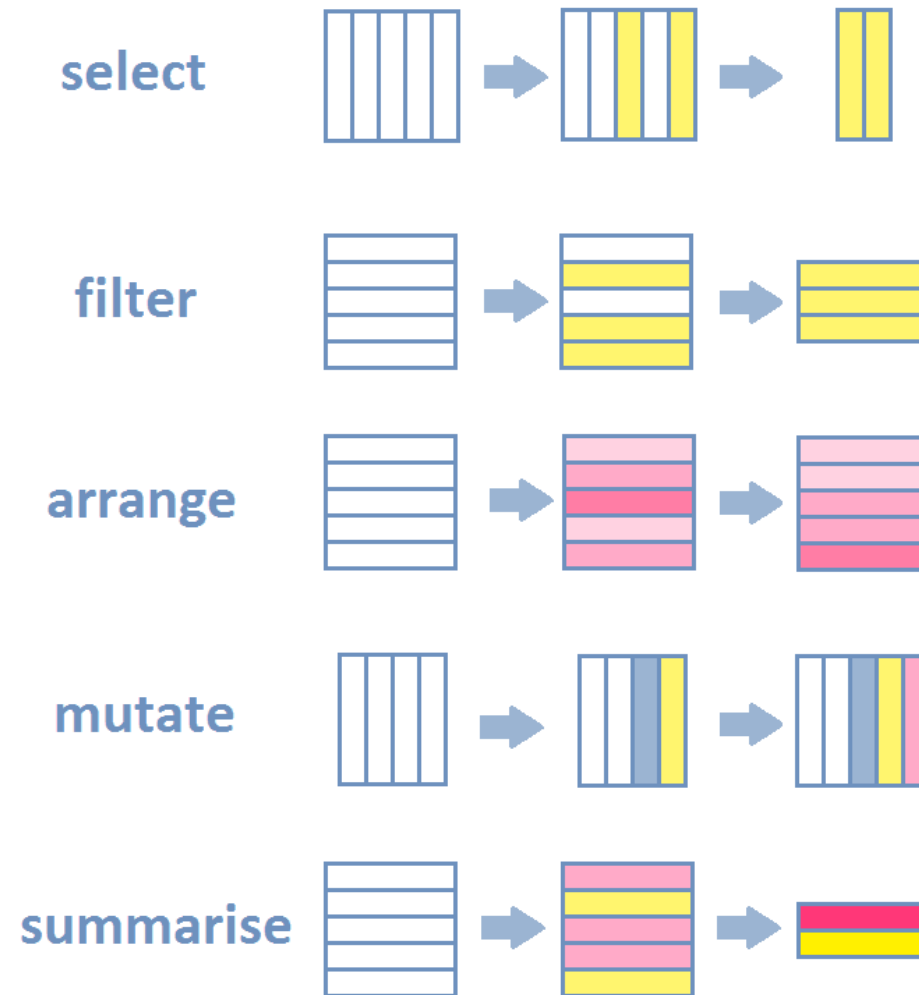
Lecture 10: Exploratory Data Analysis and Visualization, Part II

Dr. Aurélien Sallin

2025-11-12

Recap: Data cleaning and Data Visualization

Use the five building block from **dp1yr()**



Important additional tools

- `ifelse(test, yes, no)` returns a value with the same shape as the logical test. Filled with elements selected from either `yes` or `no` depending on whether the element of `test` is `TRUE` or `FALSE`

```
df |>
  mutate(gender = ifelse(gender == "m", 1, 0))
```

- `case_when` vectorises multiple `ifelse()` statements. It is the `dplyr` equivalent of `if...else`

```
df |>
  mutate(
    agegroup = case_when(
      age >= 0 & age < 19 ~ "0-18",
      age >= 19 & age < 66 ~ "19-65",
      age >= 66 & age < 100 ~ ">65",
      .default = "999"
    )
  )
```

Exploratory data analysis

“To not mislead others and not embarrass yourself, know your data”.*

- Understand the **number of observations**, the **units**, the **quality** of data, the **definitions** of variables, what to do with **missing values**.

What type of variation occurs within my variables?

- **Typical values**: mean, mode, range, standard deviation.
- **Surprising values**: outliers, rare values, unusual patterns.

What type of covariation occurs between my variables?

- **Covariation and Patterns.**

Missing Values

1. Keep in mind the **source of missingness** throughout your analysis
2. Be aware of the **trade-off between data loss and bias**
3. Choose **imputation methods wisely** (or, maybe, don't bother)



Data visualization through tables and graphs

A chart typically contains at least one axis, the values are represented in terms of visual objects (dots, lines, bars) and axes typically have scales or labels.

- If we are interested in exploring, analyzing or communicating **patterns** in the data, charts are more useful than tables.

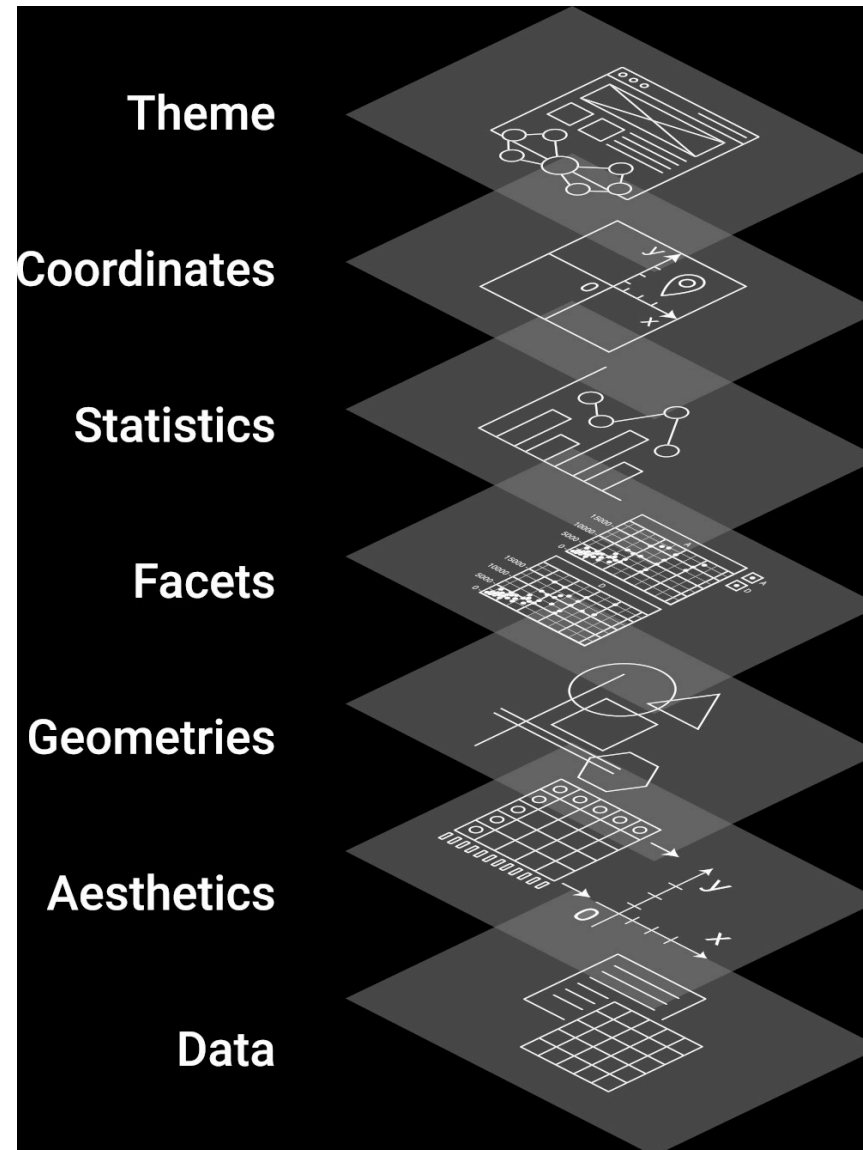
A table typically contains rows and columns, and the values are represented by text.

- If we are interested in exploring, analyzing or communicating **specific numbers** in the data, tables are more useful than graphs.

The grammar of graphics

- The *ggplot2* package is an implementation of Leland Wilkinson's 'Grammar of Graphics'.
- *ggplot2* is so good that it has become *THE* reference [In python, use *plotnine* to apply the grammar of graphics.]

Grammar of graphics with **ggplot2**



The grammar of graphics in action

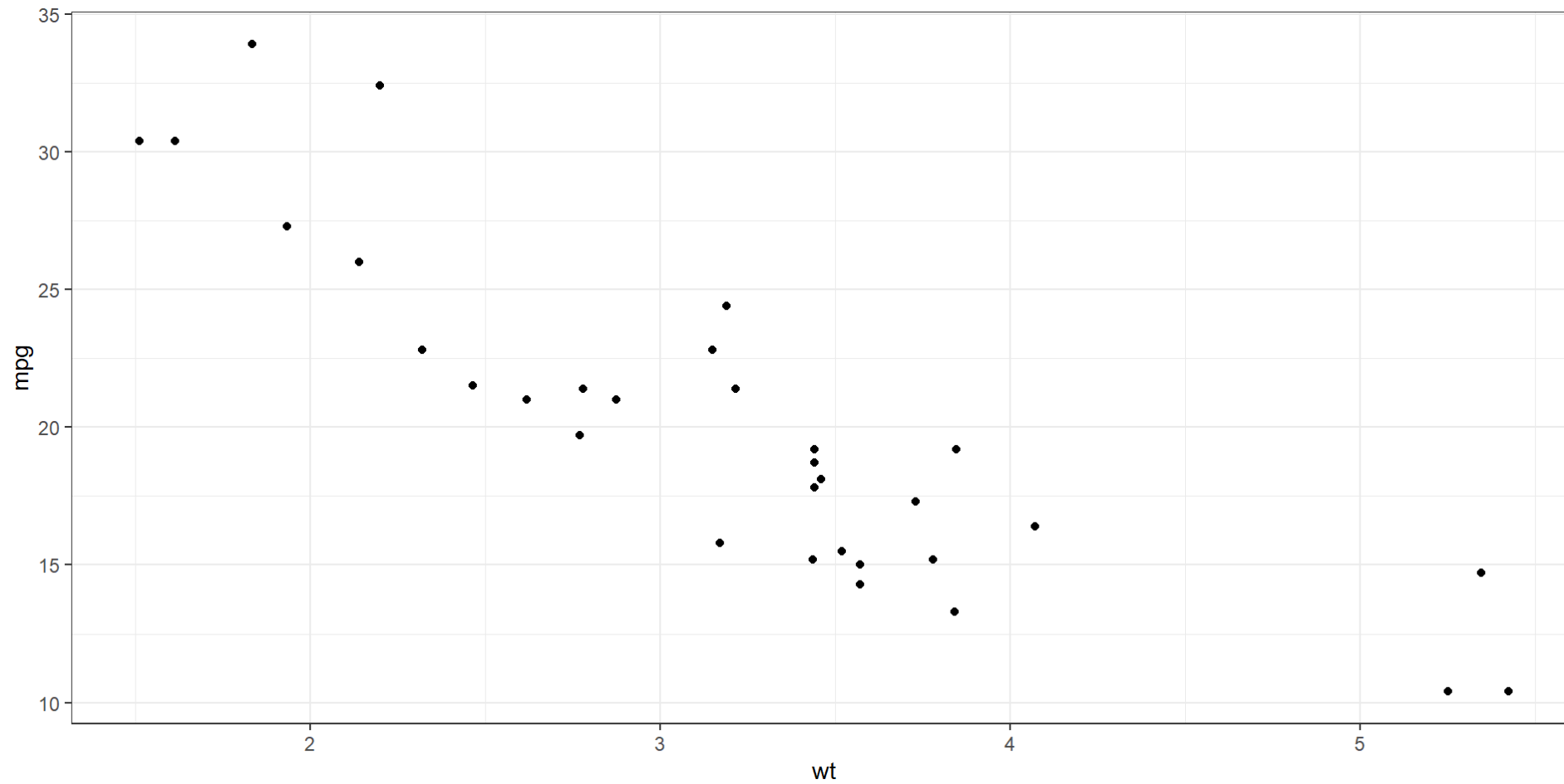
Example from [A Comprehensive Guide to the Grammar of Graphics for Effective Visualization of Multi-dimensional Data](#) using the built-in `mtcars` dataset in R.

```
mtcars # mtcars is a built-in dataset in R
```

	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
Mazda RX4	21.0	6	160.0	110	3.90	2.620	16.46	0	1	4	4
Mazda RX4 Wag	21.0	6	160.0	110	3.90	2.875	17.02	0	1	4	4
Datsun 710	22.8	4	108.0	93	3.85	2.320	18.61	1	1	4	1
Hornet 4 Drive	21.4	6	258.0	110	3.08	3.215	19.44	1	0	3	1
Hornet Sportabout	18.7	8	360.0	175	3.15	3.440	17.02	0	0	3	2
Valiant	18.1	6	225.0	105	2.76	3.460	20.22	1	0	3	1
Duster 360	14.3	8	360.0	245	3.21	3.570	15.84	0	0	3	4
Merc 240D	24.4	4	146.7	62	3.69	3.190	20.00	1	0	4	2
Merc 230	22.8	4	140.8	95	3.92	3.150	22.90	1	0	4	2
Merc 280	19.2	6	167.6	123	3.92	3.440	18.30	1	0	4	4
Merc 280C	17.8	6	167.6	123	3.92	3.440	18.90	1	0	4	4
Merc 450SE	16.4	8	275.8	180	3.07	4.070	17.40	0	0	3	3
Merc 450SL	17.3	8	275.8	180	3.07	3.730	17.60	0	0	3	3
Merc 450SLC	15.2	8	275.8	180	3.07	3.780	18.00	0	0	3	3
Cadillac Fleetwood	10.4	8	472.0	205	2.93	5.250	17.98	0	0	3	4
Lincoln Continental	10.4	8	460.0	215	3.00	5.424	17.82	0	0	3	4
Chrysler Imperial	14.7	8	440.0	230	3.23	5.345	17.42	0	0	3	4
Fiat 128	32.4	4	78.7	66	4.08	2.200	19.47	1	1	4	1
Honda Civic	30.4	4	75.7	52	4.93	1.615	18.52	1	1	4	2
Toyota Corolla	33.9	4	71.1	65	4.22	1.835	19.90	1	1	4	1
Toyota Corona	21.5	4	120.1	97	3.70	2.465	20.01	1	0	3	1
Dodge Challenger	15.5	8	318.0	150	2.76	3.520	16.87	0	0	3	2

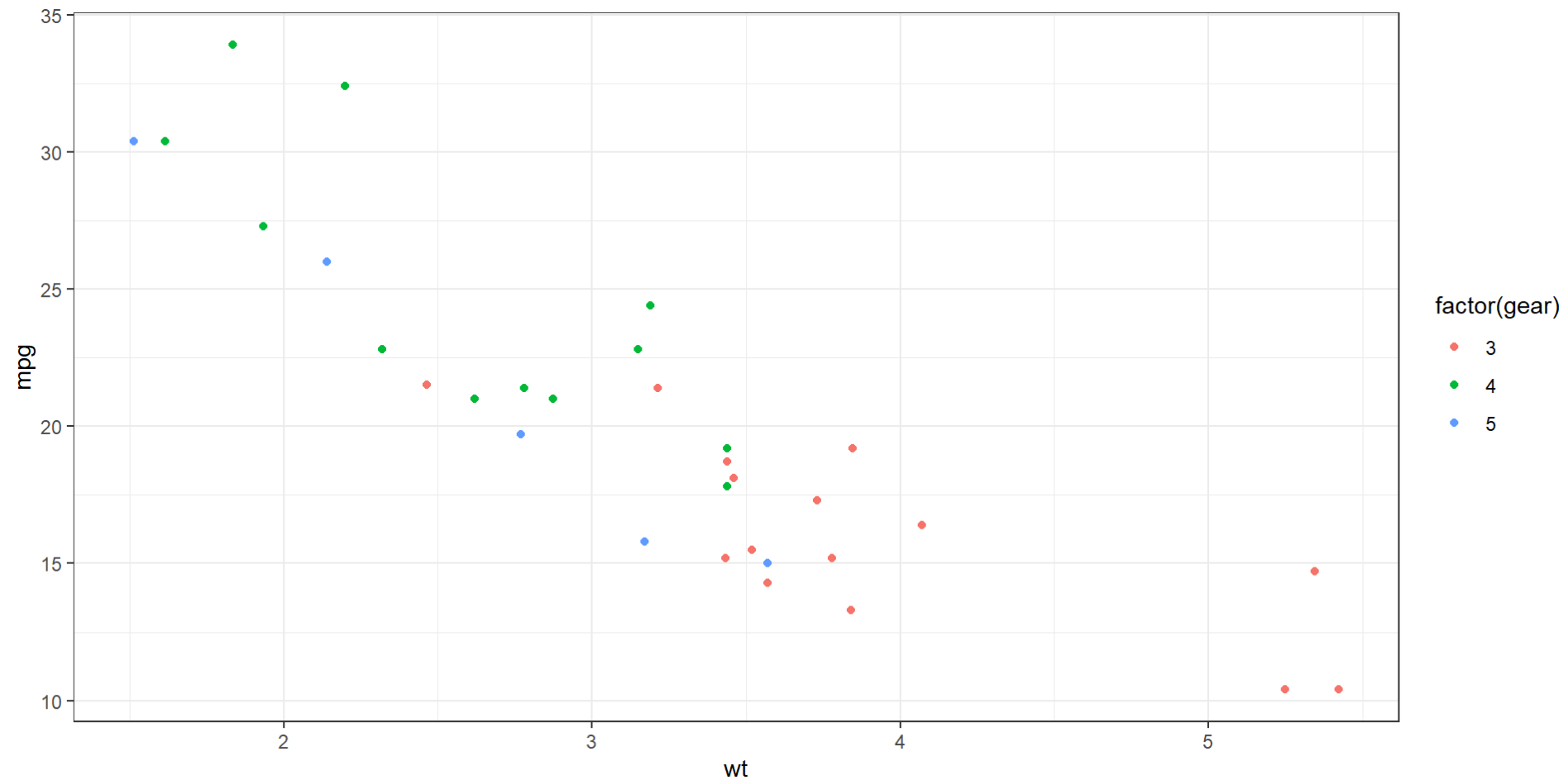
From two dimensions...

```
ggplot(mtcars, aes(x = wt, y = mpg)) +  
  geom_point() +  
  theme_bw()
```



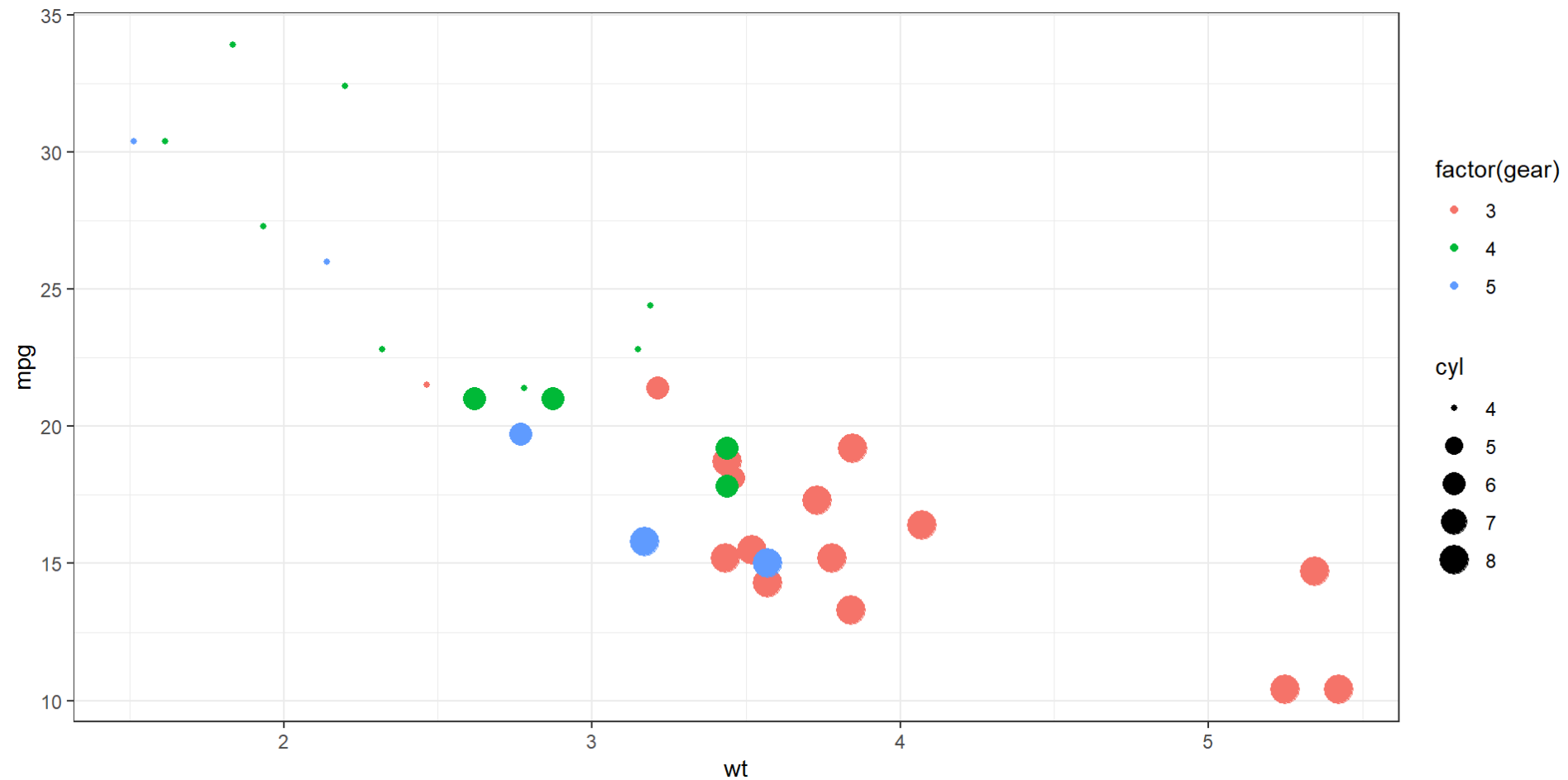
To three dimensions...

```
ggplot(mtcars, aes(x = wt, y = mpg, color=factor(gear))) +  
  geom_point() +  
  theme_bw()
```



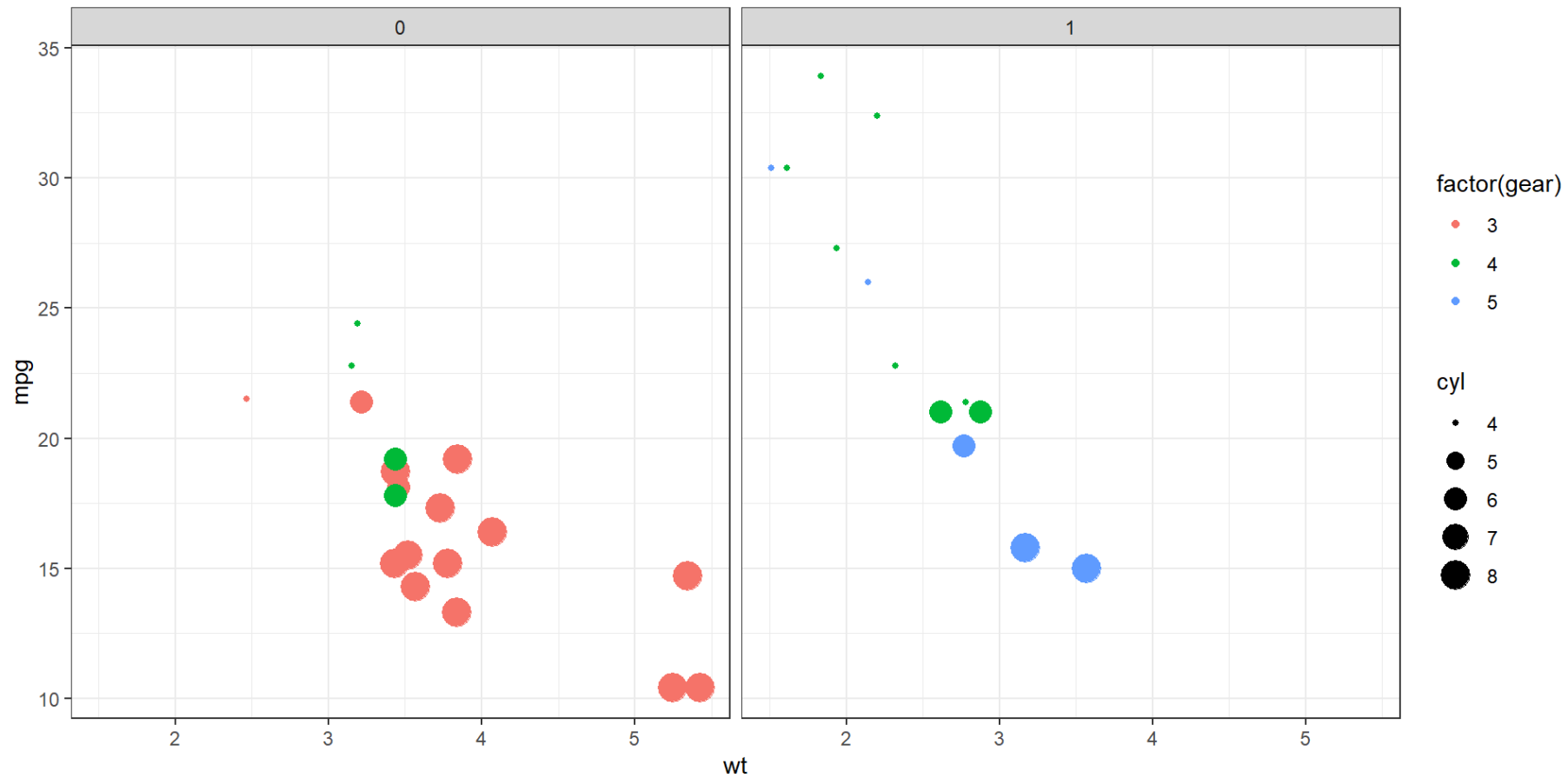
To four dimensions...

```
ggplot(mtcars, aes(x = wt, y = mpg, color=factor(gear), size = cyl)) +  
  geom_point() +  
  theme_bw()
```



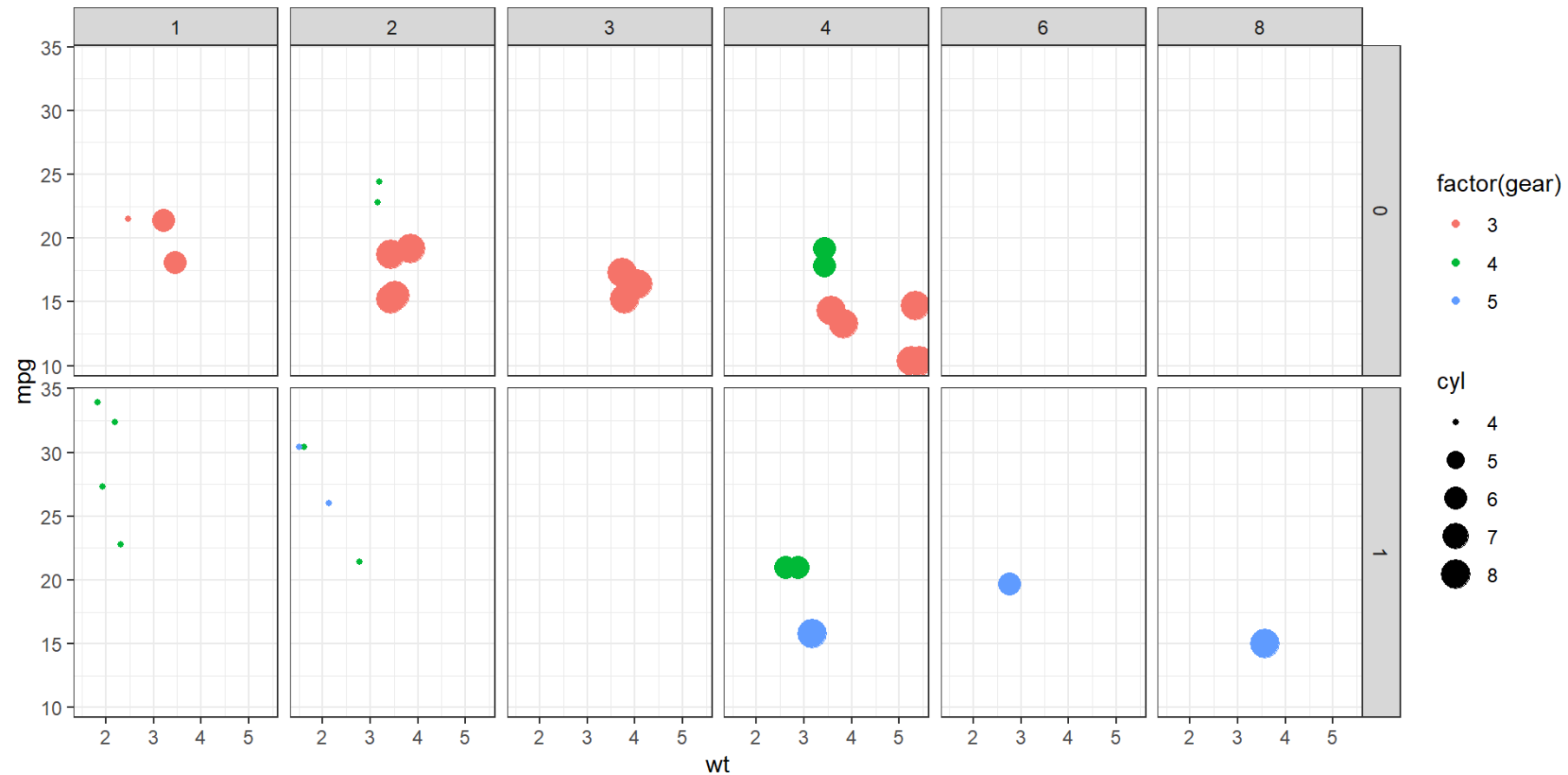
To five dimensions

```
ggplot(mtcars, aes(x = wt, y = mpg, color=factor(gear), size = cyl)) +  
  geom_point() +  
  facet_wrap(~am) +  
  theme_bw()
```



To six dimensions

```
ggplot(mtcars, aes(x = wt, y = mpg, color=factor(gear), size = cyl)) +  
  geom_point() +  
  facet_grid(am ~ carb) +  
  theme_bw()
```



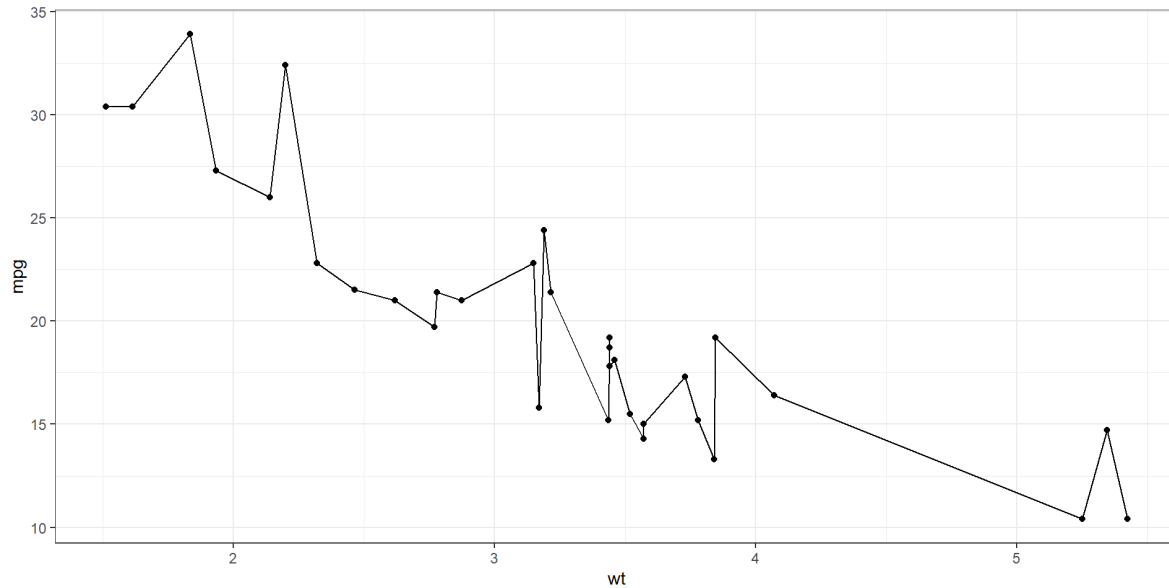
Quiz on `geom_line()` vs `geom_smooth()`

```
ggplot(mtcars, aes(x = wt, y = mpg)) +  
  geom_point() +  
  geom_line() +  
  theme_bw()
```

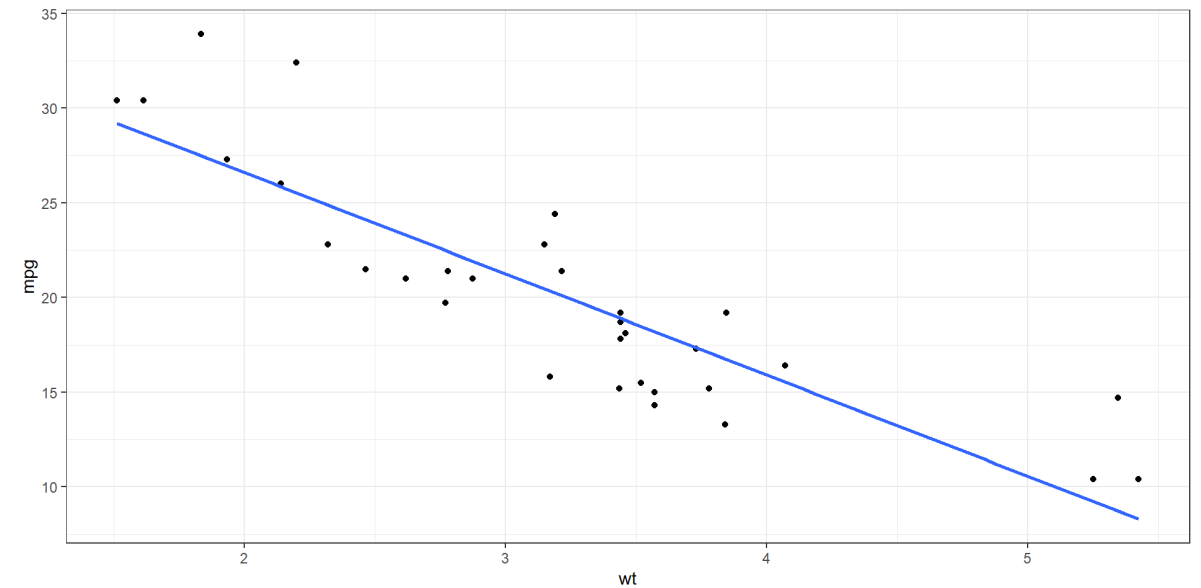
```
ggplot(mtcars, aes(x = wt, y = mpg)) +  
  geom_point() +  
  geom_smooth(method = "lm", se = FALSE) +  
  theme_bw()
```


Quiz on `geom_line()` vs `geom_smooth()`

```
ggplot(mtcars, aes(x = wt, y = mpg)) +  
  geom_point() +  
  geom_line() +  
  theme_bw()
```



```
ggplot(mtcars, aes(x = wt, y = mpg)) +  
  geom_point() +  
  geom_smooth(method = "lm", se = FALSE) +  
  theme_bw()
```



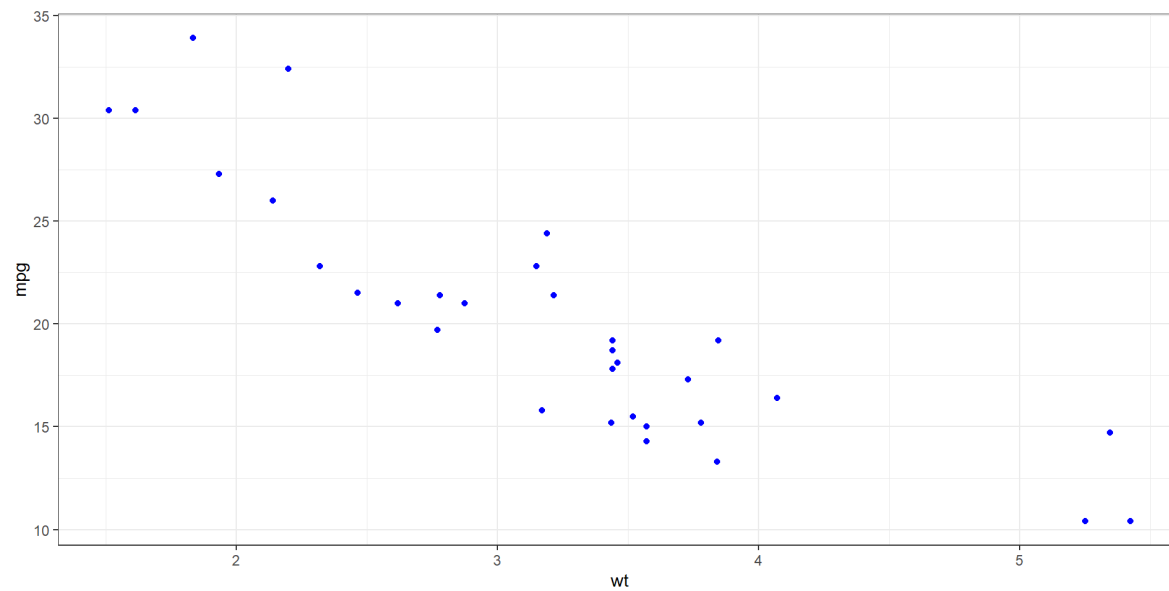
Quiz on size, fill, color arguments

```
mtcars |>
  ggplot(aes(x = wt, y = mpg)) +
  geom_point(color = "blue") +
  theme_bw()
```

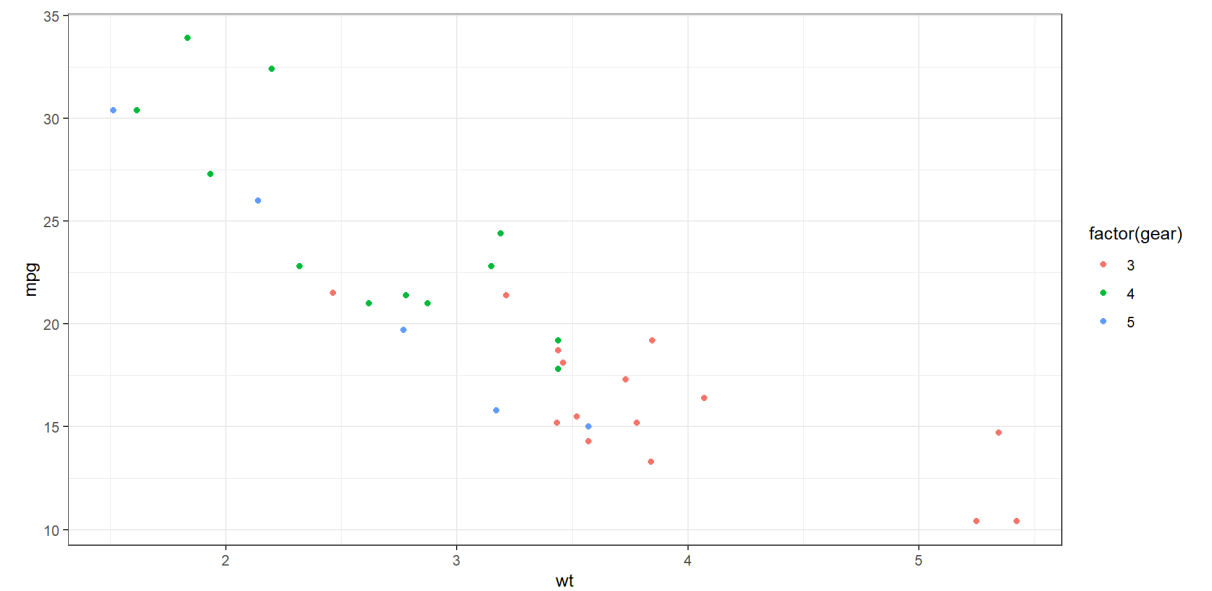
```
mtcars |>
  ggplot(aes(x = wt, y = mpg, color=factor(gear))) +
  geom_point() +
  theme_bw()
```

Quiz on size, fill, color arguments

```
mtcars |>
  ggplot(aes(x = wt, y = mpg)) +
  geom_point(color = "blue") +
  theme_bw()
```



```
mtcars |>
  ggplot(aes(x = wt, y = mpg, color=factor(gear))) +
  geom_point() +
  theme_bw()
```



Quiz on `geom_bar()`

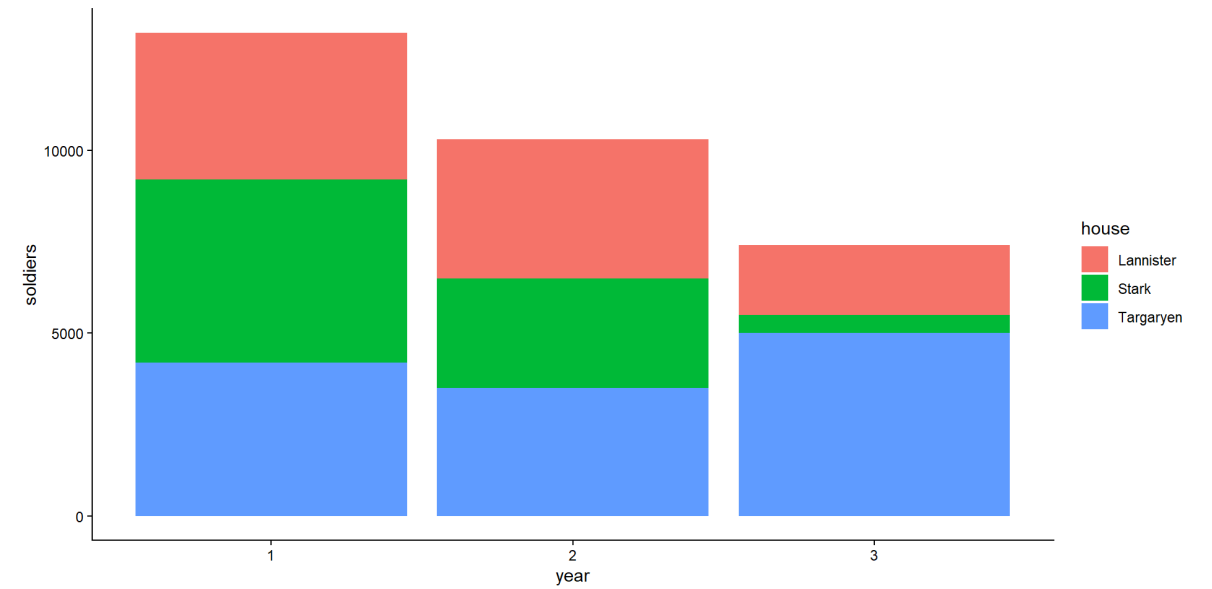
Which code produced the figure? (This question would not be an exam question, as it requires specific knowledge of `geom_bar()`. Solve it with R.)

```
got_data <- tibble(  
  house = c("Stark", "Stark", "Stark", "Lannister", "Lannister", "Lannister",  
            "Targaryen", "Targaryen", "Targaryen"),  
  soldiers = c(5000, 3000, 500, 4000, 3800, 1900, 4200, 3500, 5000),  
  year = c(1,2,3,1,2,3,1,2,3)  
)
```

```
ggplot(got_data, aes(x = year, y = soldiers, fill = house)) +  
  geom_bar(stat = "identity") +  
  theme_classic()
```

```
ggplot(got_data, aes(x = year, y = soldiers, fill = house)) +  
  geom_histogram() +  
  theme_classic()
```

```
ggplot(got_data, aes(x = year, fill = house)) +  
  geom_bar() +  
  theme_classic()
```



Today

Goals for today

Building on what we covered last week:

1. Know how to conduct exploratory data analysis (EDA).
2. Visualize data using tables.
3. Visualize data using the grammar of graphics.
4. Produce effective data visualization.

From graphs to effective data visualization

Data visualization: some principles

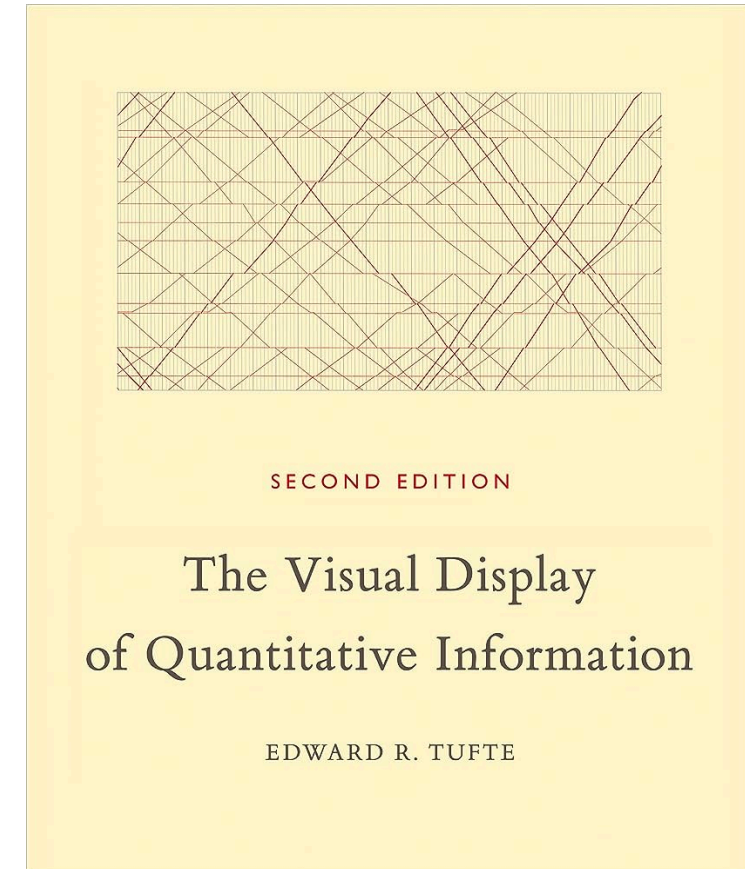
- Values are represented by their **position relative to the axes**: line charts and scatterplots.
- Values are represented by the **size of an area**: bar charts and area charts.
- Values are **continuous**: use chart type that visually connects elements (line chart).
- Values are **categorical**: use chart type that visually separates elements (bar chart).

“Greatest number of ideas in
the shortest time with the
least ink in the smallest space”
(Edward Tufte, 1983)

Data visualization: some principles

Recommendations from Edward Tufte's "The Visual Display of Quantitative Information" (1983)

1. Lie Factor
2. Data-to-ink Ratio
3. Data density

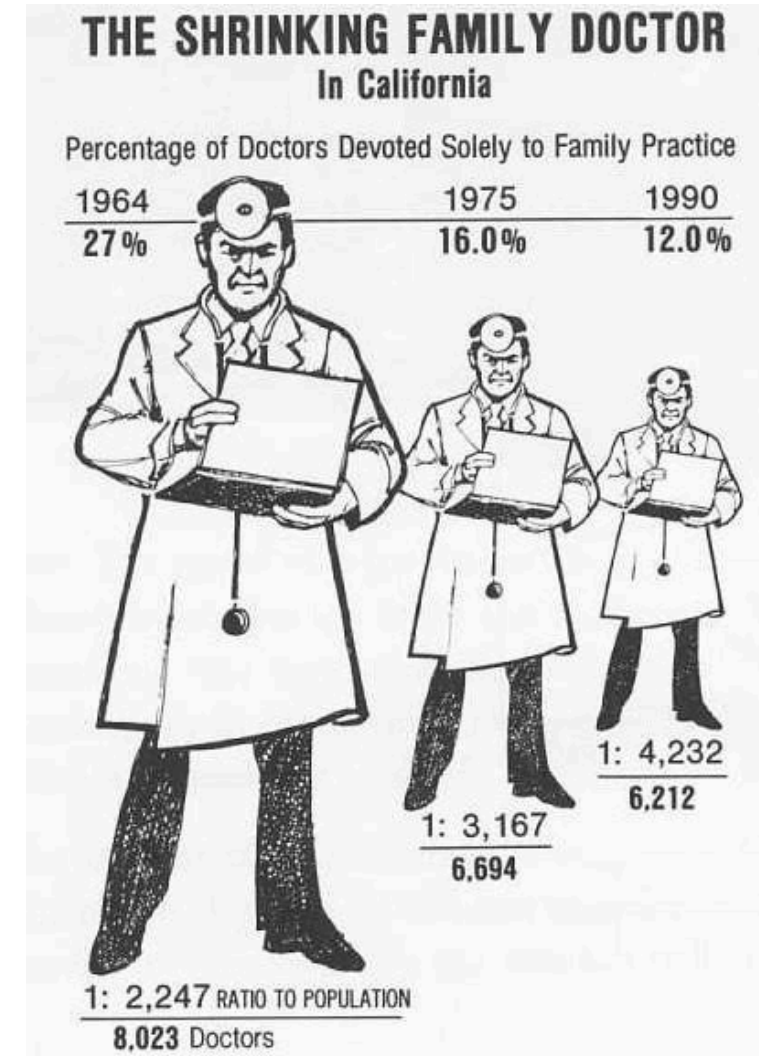


Lie Factor, or strive for graphical integrity

We can quantify the **Lie Factor** of a graph as a measure of how much the graphic distorts the data.

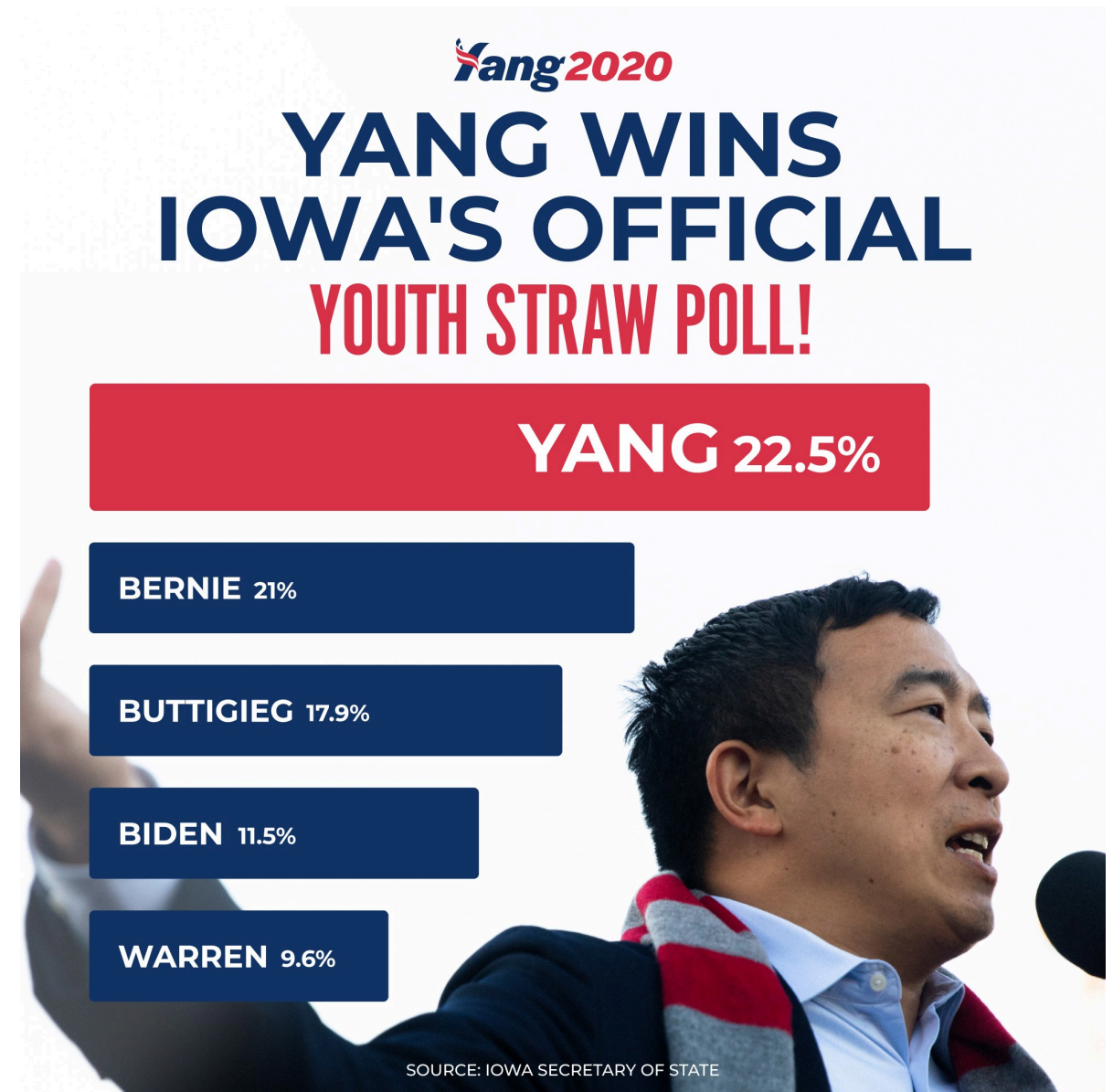
“The representation of numbers, as physically measured on the surface of the graphic itself, should be directly proportional to the quantities represented.” (Tufte, 1983)

$$\text{Lie Factor} = \frac{\text{size of effect shown in graphic}}{\text{size of effect in data}}$$



Lie Factor, or strive for graphical integrity

$$\text{Lie Factor} = \frac{\text{Yang had 39.1\% of total ink}}{\text{Yang had 22.5\%}} = 1.74$$



Thou shalt not truncate the Y axis.

Tucker Carlson is guilty of committing chart sins

Thou shalt not truncate the Y axis.

JEMIMA KELLY

+ Add to myFT



Jemima Kelly SEPTEMBER 28 2021

48

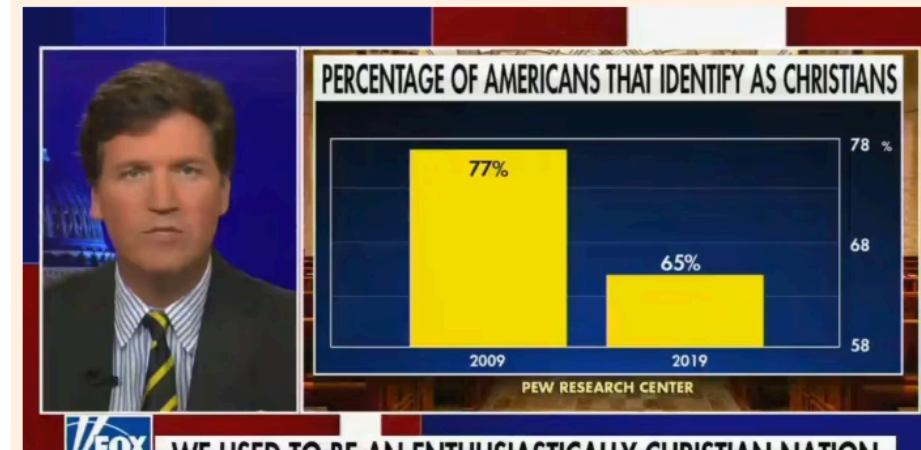
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True chart crime fans, rejoice! For unto you a new atrocity is born:



Thou shalt not truncate the Y axis.

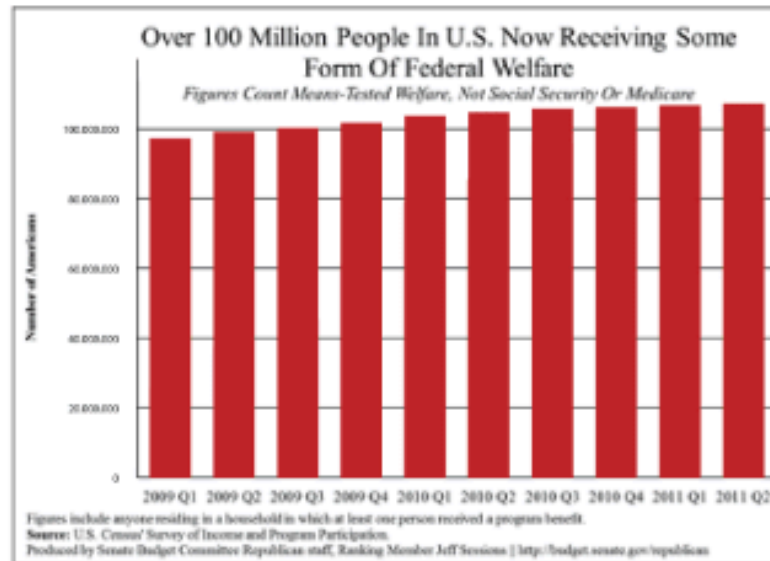
THE BLOG

Over 100 Million Now Receiving Federal Welfare

2:40 PM, AUG 8, 2012 • BY DANIEL HALPER

DETAILS PAGE PRINT LARGER TEXT A+ SPREADER TEXT A- COMMENTS

A new chart set to be released later today by the Republican side of the Senate Budget Committee details a startling statistic: "Over 100 Million People in U.S. Now Receiving Some Form Of Federal Welfare."



lie factor: 1

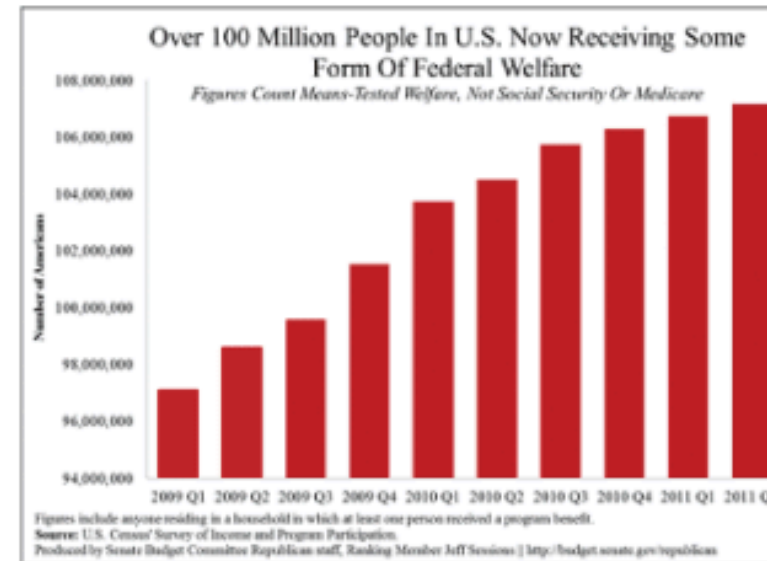
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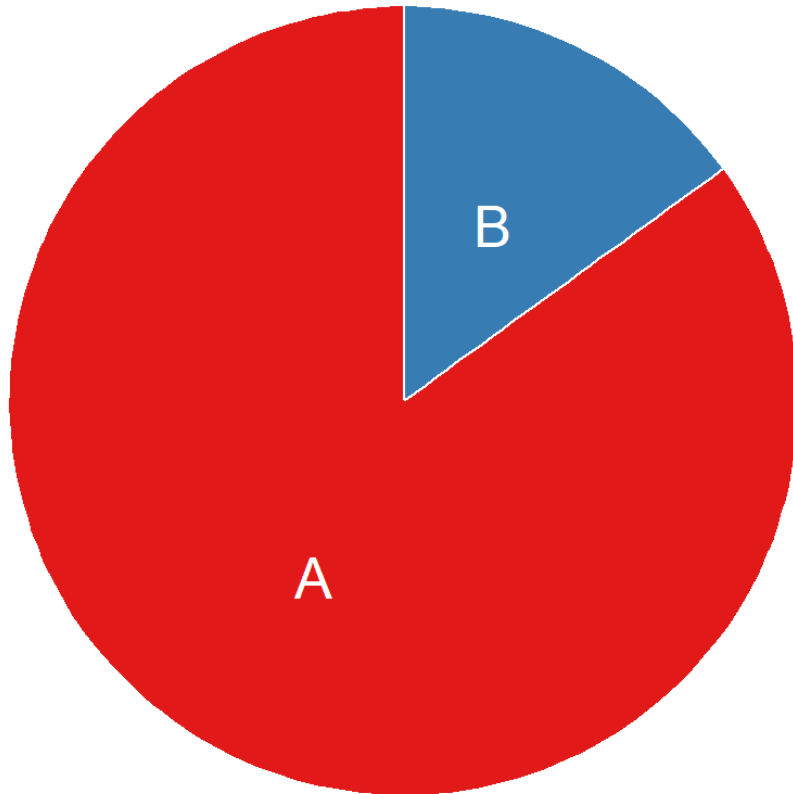


lie factor: 16,08

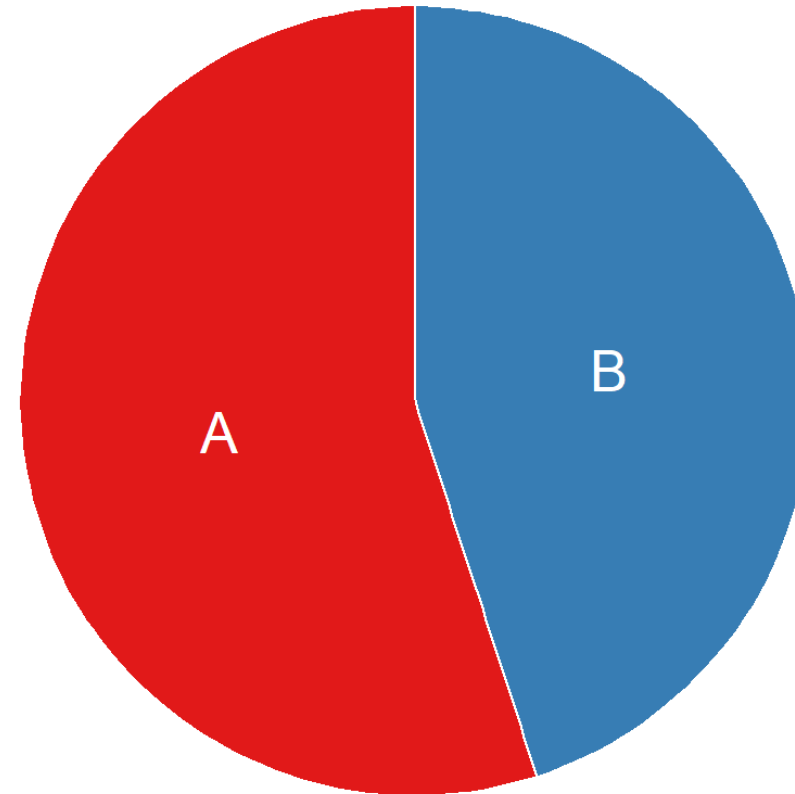
Source: The lie factor and the baseline paradox

Avoid pie charts

Pie Chart



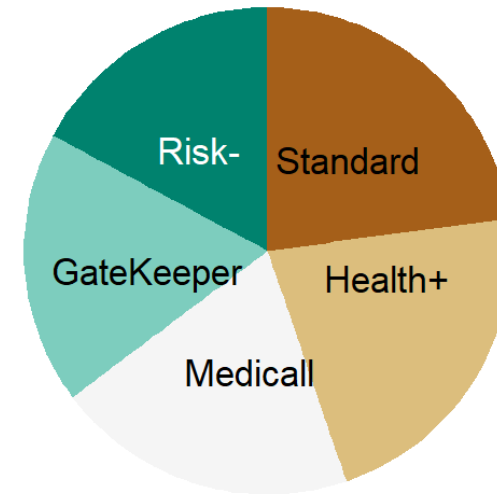
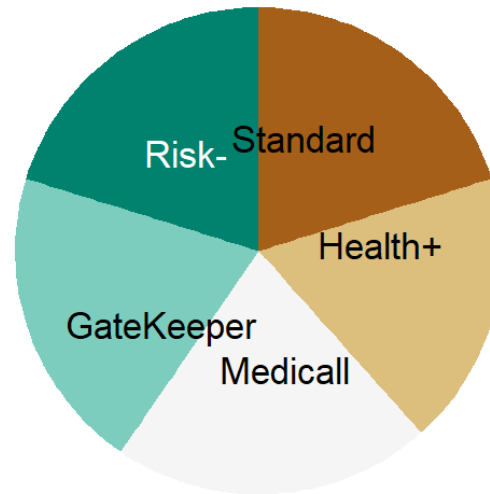
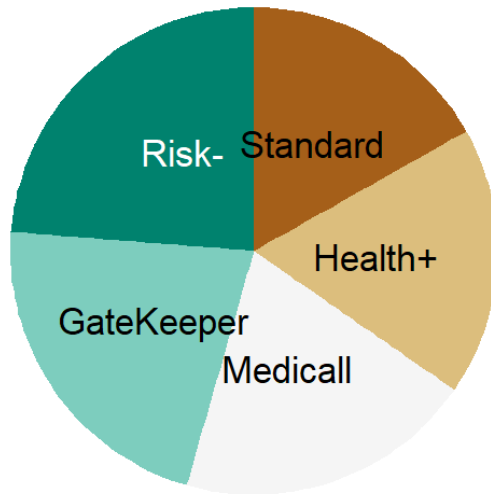
Pie Chart



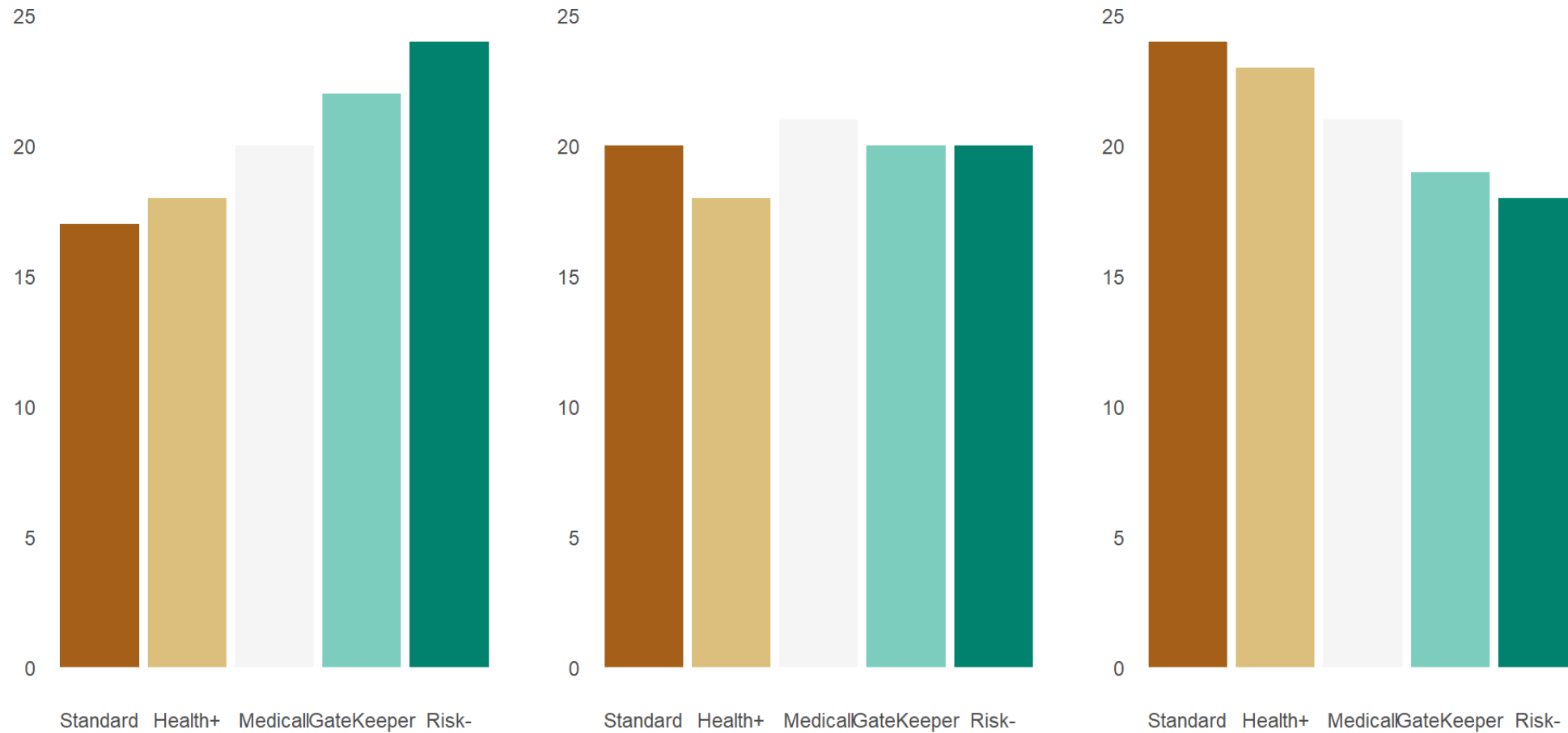
“All variations lead to overestimation of small values and underestimation of large ones.” [Kosara et al, 2018](#)

Avoid pie charts, please!

► Show the code



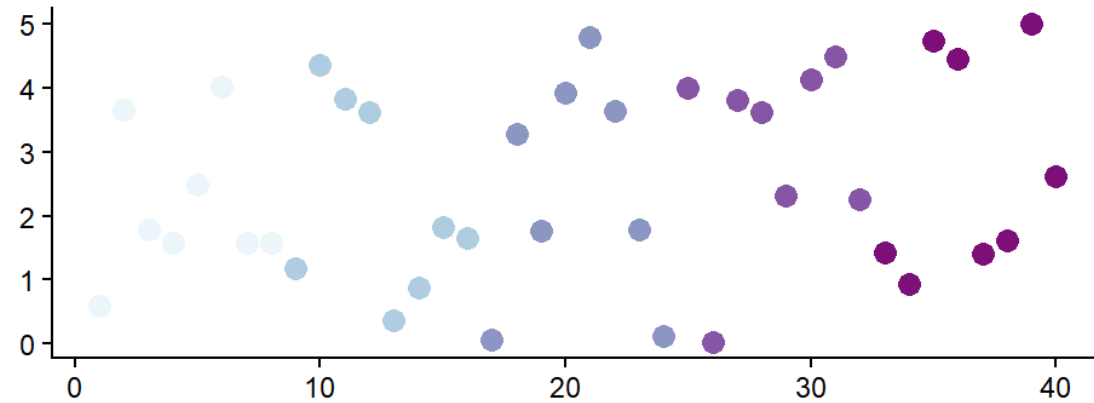
Avoid pie charts, please!



Different types of colors for different types of data

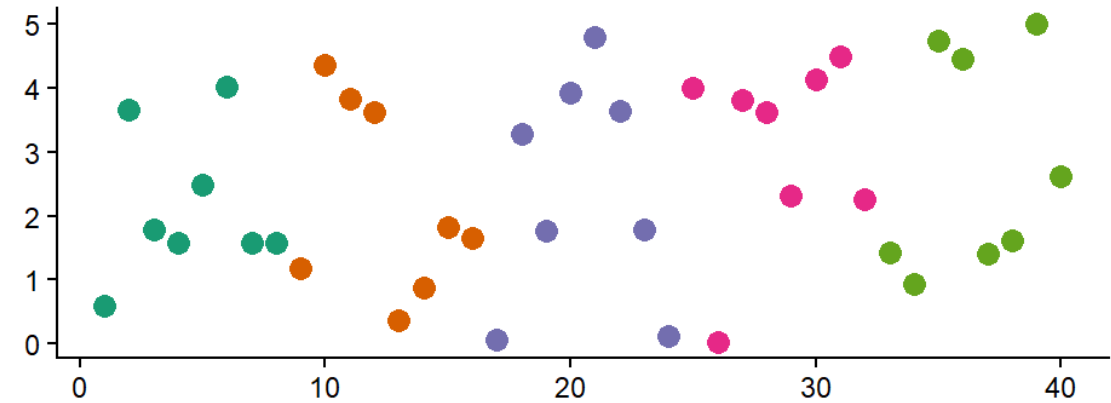
Palette: Sequential

`scale_color_brewer(palette = "BuPu")`



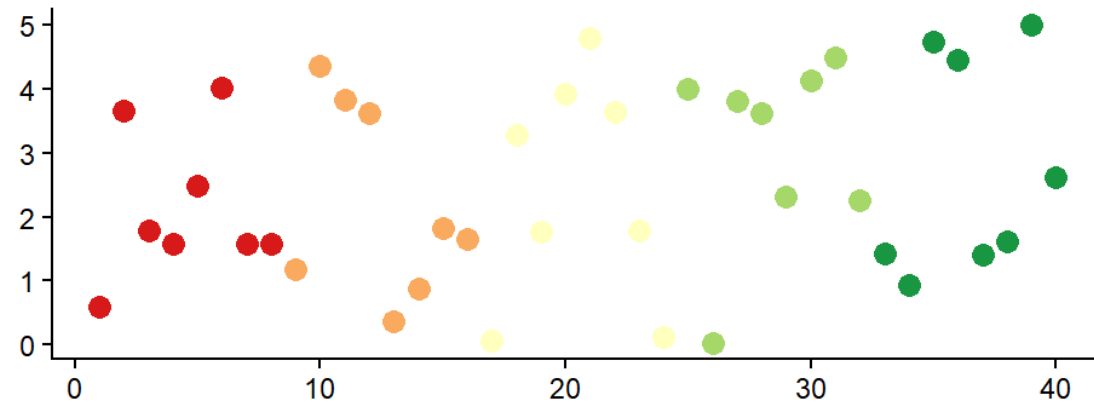
Palette: Qualitative

`scale_color_brewer(palette = "Dark2")`



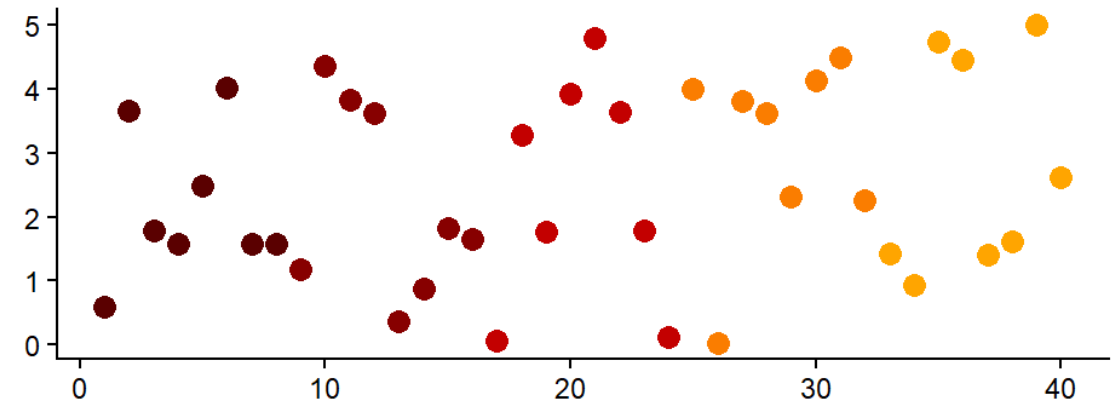
Palette: R Color brewer: Diverging

`scale_color_brewer(palette = "RdYlGn")`



Palette: Gryffindor 🐉

`scale_color_hp(house = "Gryffindor", discrete = TRUE)`



Only what matters should be reported

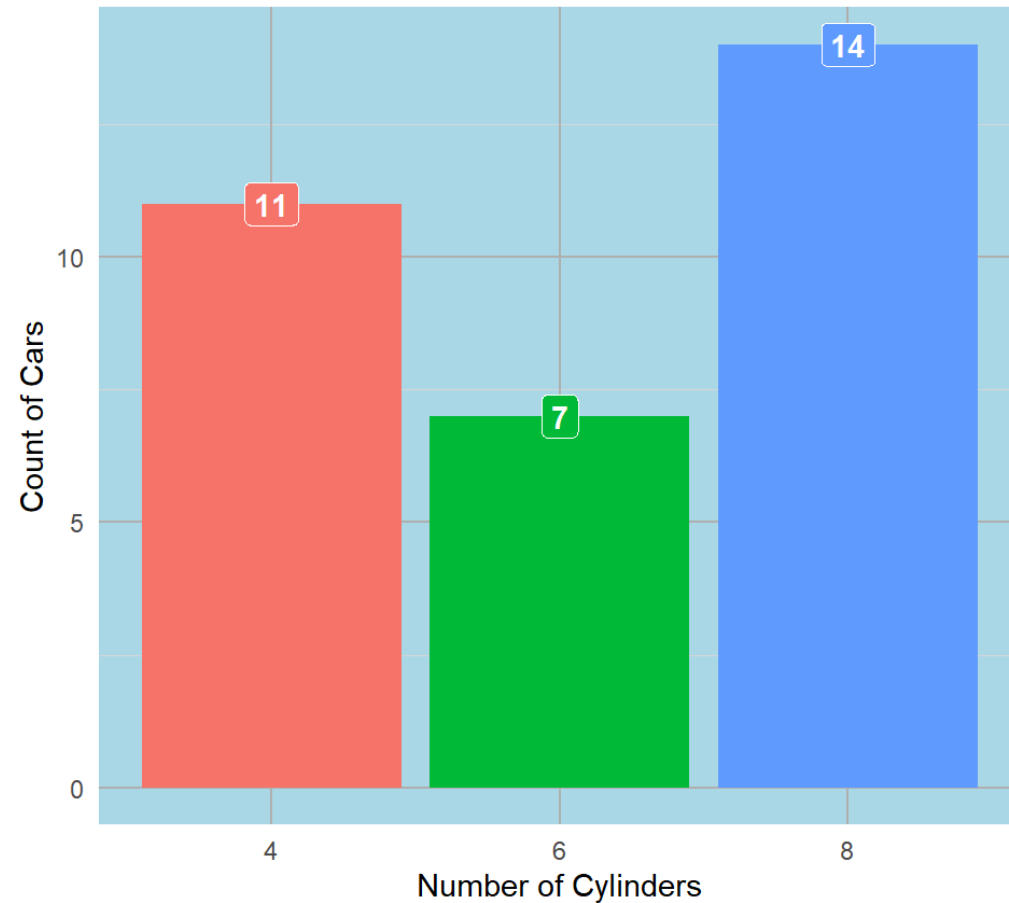
- Data-ink Ratio = $\frac{\text{ink used for data points}}{\text{total ink used to print the graphic}}$
 - Proportion of a graphic's ink devoted to the non-redundant display of data-information
 - **Data ink:** data points and measured quantities, such as the dots in a scatter plot
 - **Non-data ink:** functional marks such as titles, labels, axes, gridlines and tick points or decorative marks

Idea: maximize data-ink ratio

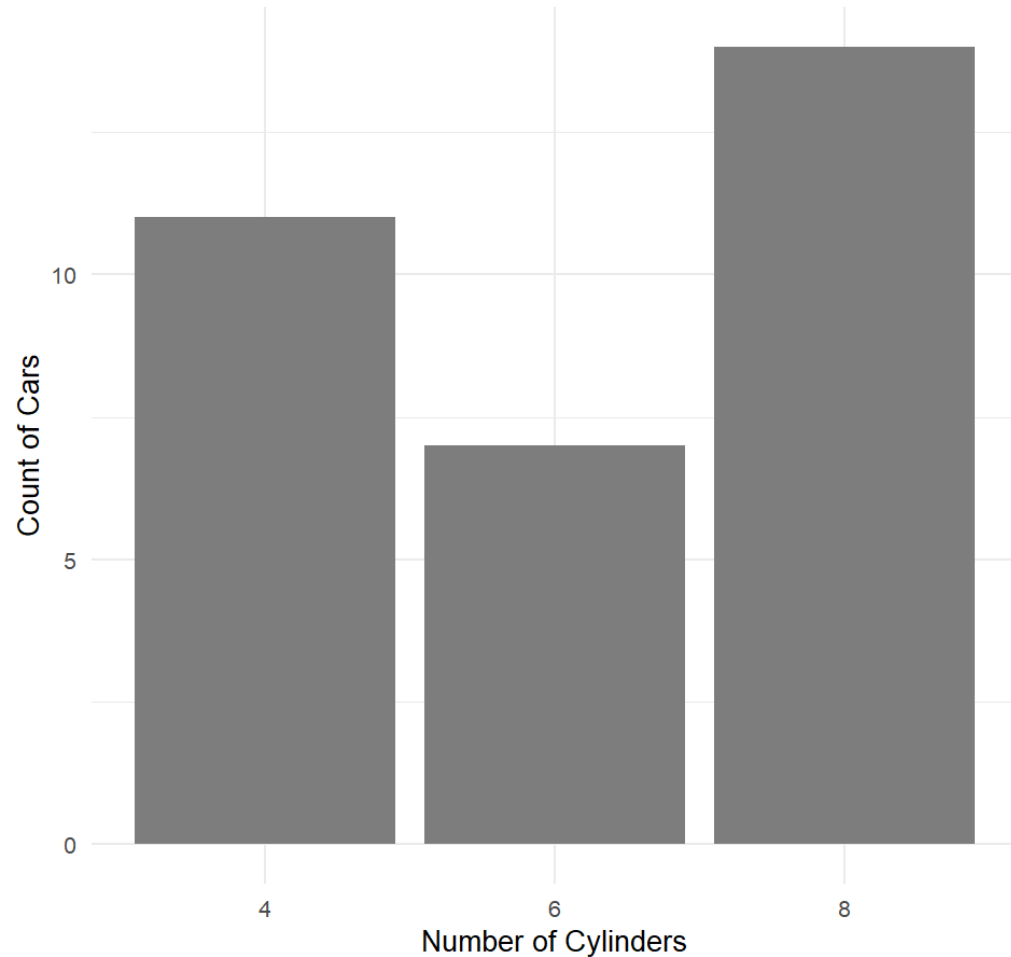
- Limits to this approach: we still need some ink to interpret and understand the data.
- No chart junk!

Only what matters should be reported

Car Cylinder Count with Low Data-Ink Ratio
Detailed representation with color, labels, and gridlines

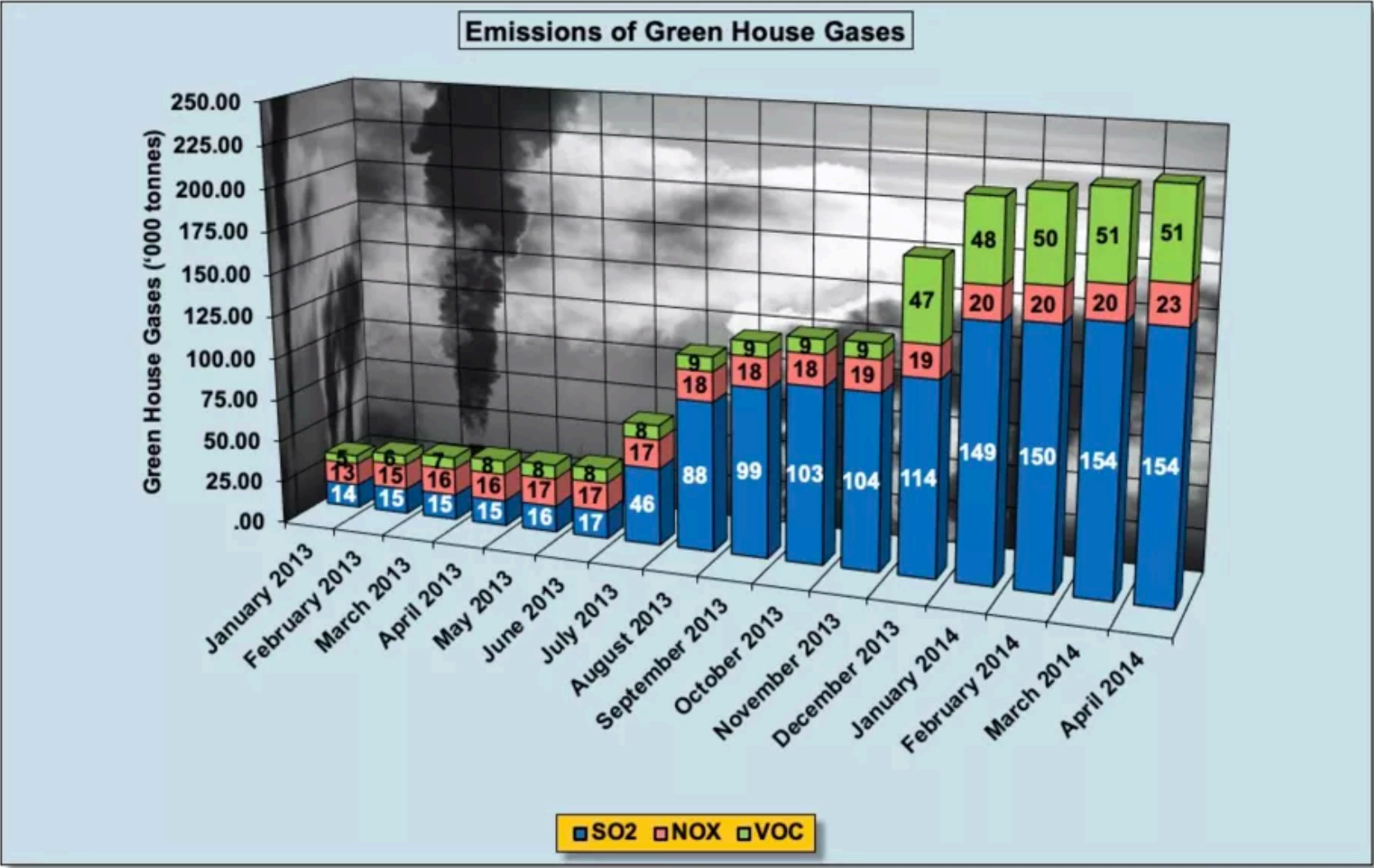


Car Cylinder Count with High Data-Ink Ratio



► Show code for the graphs

Only what matters should be reported

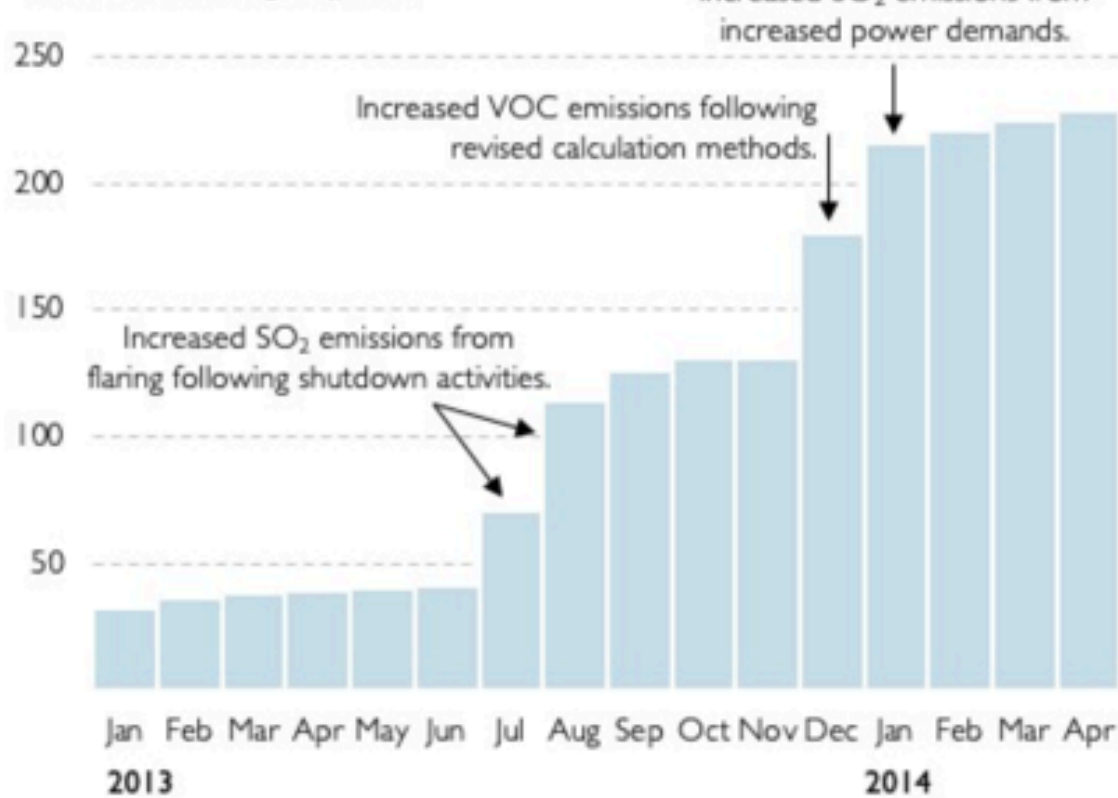


Source: simplexct.com

Only what matters should be reported

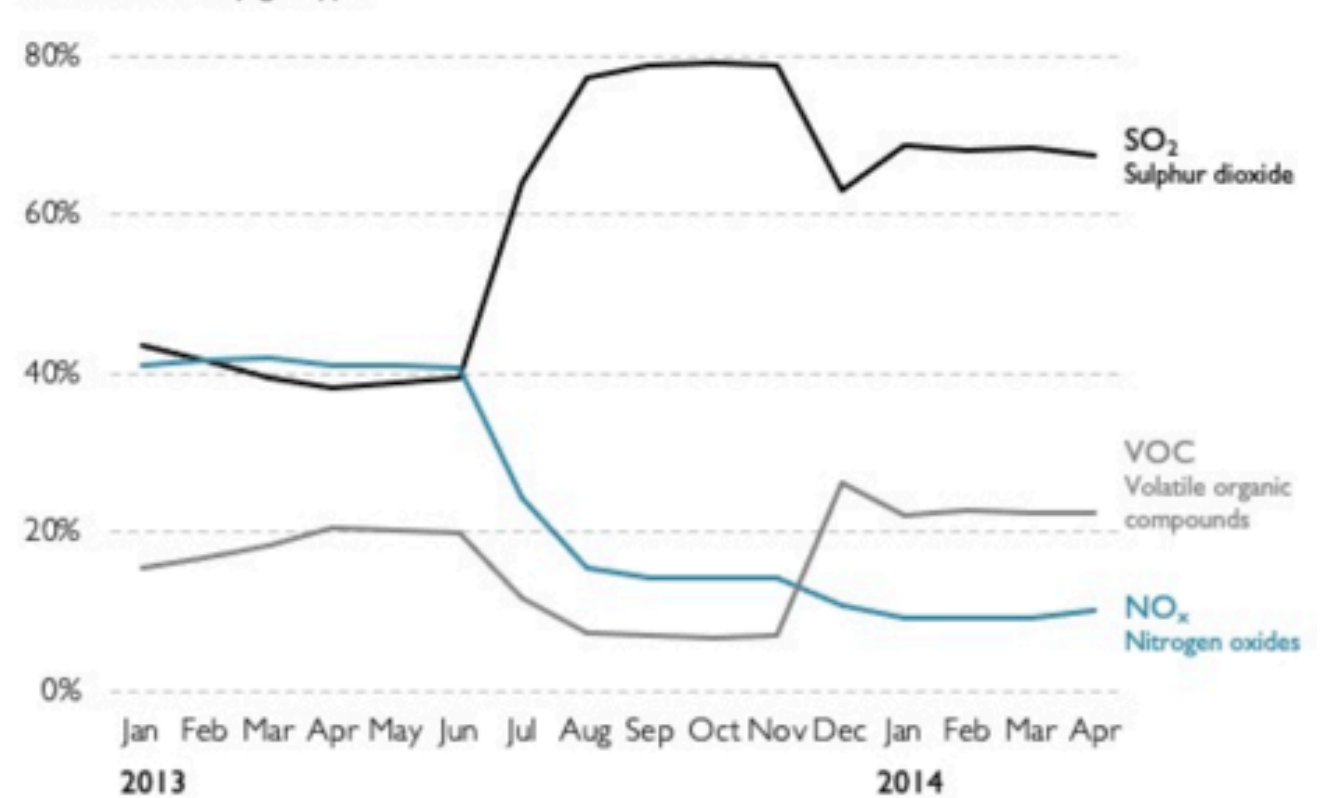
Emissions of Green House Gases

'000 Tonnes of SO₂/NO_x/VOC



Emissions of Green House Gases

Share of total , by gas type



Source: simplexct.com

Only what matters should be reported

Works for tables as well...

Short-term credit lines by bank

June 30, 1980 (millions USD)

	Finance Companies					Manufacturing Companies							
Bank	Canada	Germany	UK	USA	Total	Argentina	Australia	Canada	France	UK	USA	Total	Grand Total
Allied & Associates	0.0	0.0	18.6	0.0	18.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	18.6
Bank of America	0.0	0.0	0.0	17.5	17.5	0.0	3.3	0.0	0.0	0.0	17.5	20.8	38.3
Bankers Trust	0.0	0.0	0.0	13.0	13.0	0.0	0.0	0.0	0.0	0.0	13.0	13.0	26.0
Banque National de Paris	0.0	0.0	11.3	0.0	11.3	0.0	0.0	15.0	34.6	0.0	0.0	49.6	60.9
Barclays	0.0	0.0	42.5	0.0	42.5	0.0	0.0	0.0	0.0	133.4	0.0	133.4	175.9
Chase Manhattan	0.0	0.0	0.0	15.2	15.2	0.0	0.0	0.0	0.0	0.0	15.0	15.0	30.2
Chemical	0.0	0.0	0.0	13.3	13.3	0.0	0.0	0.0	0.0	0.0	13.0	13.0	26.3
CIBC	37.8	0.0	27.1	0.0	64.9	0.0	0.0	222.9	3.6	3.4	0.0	229.9	294.8
Citibank	0.0	0.0	0.0	17.5	17.5	0.5	0.0	0.0	0.0	1.1	17.5	19.1	36.6
Commerzbank	0.0	3.3	0.0	0.0	3.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	3.3
Continental Illinois	0.0	0.0	4.6	21.8	26.4	0.0	0.0	0.0	0.0	0.0	21.1	21.1	47.5
Crédit Lyonnais	0.0	0.0	9.0	0.0	9.0	0.0	0.0	0.0	33.0	0.0	3.0	36.0	45.0
Deutsche Bank	0.0	3.3	0.0	0.0	3.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	3.3
Dresdner	0.0	3.3	0.0	0.0	3.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	3.3
FNB Chicago	0.0	0.0	0.0	15.0	15.0	0.0	0.0	0.0	0.0	0.0	15.0	15.0	30.0
Lloyds	0.0	0.0	42.5	0.0	42.5	0.0	0.0	0.0	0.0	36.2	0.0	36.2	78.7
Midland	0.0	0.0	54.3	0.0	54.3	0.0	0.0	0.0	0.0	36.2	0.0	36.2	90.5
Royal Bank of Canada	0.0	0.0	11.3	0.0	11.3	0.0	0.0	0.0	0.0	0.0	15.0	15.0	26.3
Société General	0.0	0.0	9.0	0.0	9.0	0.0	0.0	0.0	50.1	0.0	0.0	50.1	59.1
Toronto Dominion	0.0	0.0	6.8	0.0	6.8	0.0	0.0	0.0	0.0	0.0	15.2	15.2	22.0
Others	0.0	7.8	50.8	78.1	136.7	40.1	25.5	0.0	16.1	19.2	48.2	149.1	285.8
Total	37.8	17.7	287.8	191.4	534.7	40.6	28.8	237.9	137.4	229.5	193.5	867.7	1,402.4

Short-term credit lines by bank

June 30, 1980 (millions USD)

Bank	Grand Total	MANUFACTURING COMPANIES							FINANCE COMPANIES				
		Canada	UK	USA	France	Argentina	Australia	Total	UK	USA	Canada	Germany	Total
Total	1,402.4	237.9	229.5	193.5	137.4	40.6	28.8	867.7	287.8	191.4	37.8	17.7	534.7
CIBC	294.8	222.9	3.4	-	3.6	-	-	229.9	27.1	-	37.8	-	64.9
Others	285.8	-	19.2	48.2	16.1	40.1	25.5	149.1	50.8	78.1	-	7.8	136.7
Barclays	175.9	-	133.4	-	-	-	-	133.4	42.5	-	-	-	42.5
Midland	90.5	-	36.2	-	-	-	-	36.2	54.3	-	-	-	54.3
Lloyds	78.7	-	36.2	-	-	-	-	36.2	42.5	-	-	-	42.5
Banque National de Paris	60.9	15.0	-	-	34.6	-	-	49.6	11.3	-	-	-	11.3
Société General	59.1	-	-	-	50.1	-	-	50.1	9.0	-	-	-	9.0
Continental Illinois	47.5	-	-	21.1	-	-	-	21.1	4.6	21.8	-	-	26.4
Crédit Lyonnais	45.0	-	-	3.0	33.0	-	-	36.0	9.0	-	-	-	9.0
Bank of America	38.3	-	-	17.5	-	-	3.3	20.8	-	17.5	-	-	17.5
Citibank	36.6	-	1.1	17.5	-	0.5	-	19.1	-	17.5	-	-	17.5
Chase Manhattan	30.2	-	-	15.0	-	-	-	15.0	-	15.2	-	-	15.2
FNB Chicago	30.0	-	-	15.0	-	-	-	15.0	-	15.0	-	-	15.0
Royal Bank of Canada	26.3	-	-	15.0	-	-	-	15.0	11.3	-	-	-	11.3
Chemical	26.3	-	-	13.0	-	-	-	13.0	-	13.3	-	-	13.3
Bankers Trust	26.0	-	-	13.0	-	-	-	13.0	-	13.0	-	-	13.0
Toronto Dominion	22.0	-	-	15.2	-	-	-	15.2	6.8	-	-	-	6.8
Allied & Associates	18.6	-	-	-	-	-	-	-	18.6	-	-	-	18.6
Commerzbank	3.3	-	-	-	-	-	-	-	-	-	-	3.3	3.3
Deutsche Bank	3.3	-	-	-	-	-	-	-	-	-	-	3.3	3.3
Dresdner	3.3	-	-	-	-	-	-	-	-	-	-	3.3	3.3

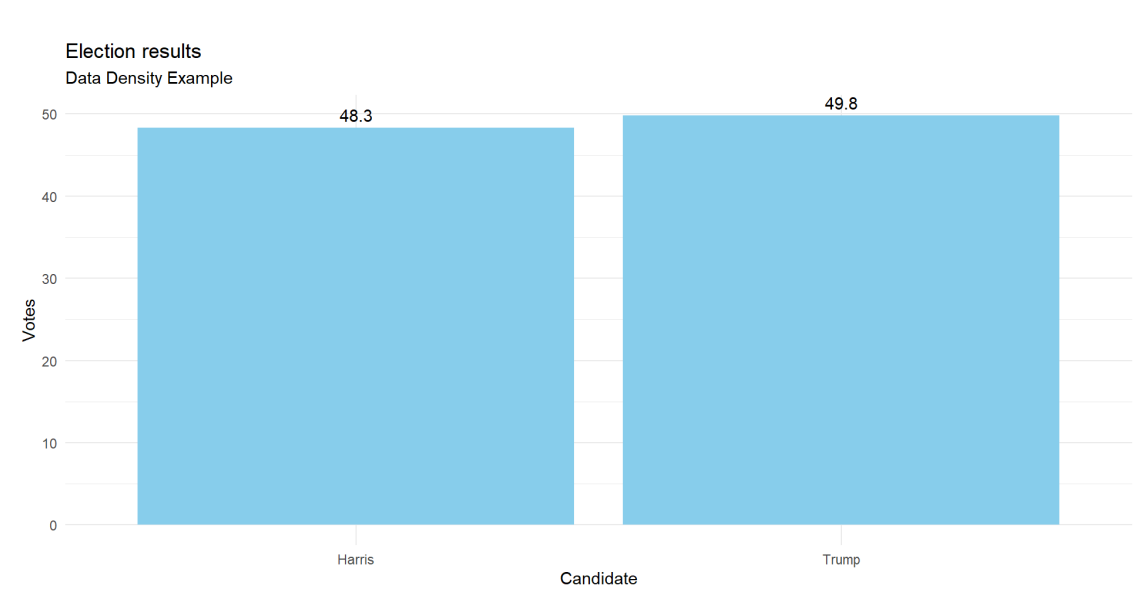
Source: [simplerxct.com](https://www.simplerxct.com)

Data density

The data density takes the number of data points that are being graphed relative to the physical size of the graphic to capture the principle of aiming to present many numbers in a small space:

- Data density = $\frac{\text{number of entries in data matrix}}{\text{area of data graphic}}$

Data density



1. Graph

Election results
Data Density Example

candidate	votes
Trump	49.8
Harris	48.3

2. Table

49.8% voted for
Trump, 48.3% for
Harris.

3. Text

Data visualization: from a graph to a story

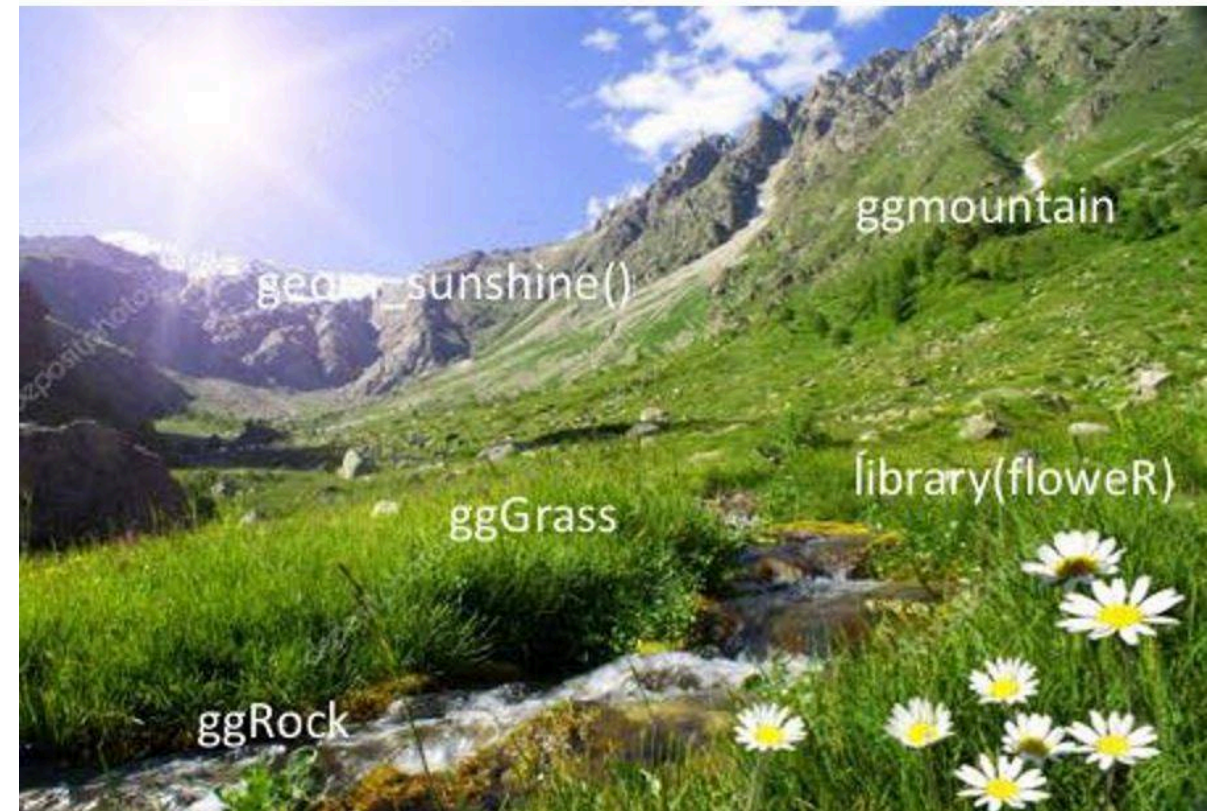
Two pieces of advice I personally received:

1. If possible, **fit your whole story in one graph.**
 2. Your audience should understand your graph **without the need of listening to you or reading your text.**
- Be simple and avoid unnecessary fanciness.
 - Avoid pie charts and 3D charts.

Have fun with ggplot

- Follow great dataviz leaders and influencers to learn how to master your craft: [data-to-viz](#) (great!), [Yan Holtz](#), [Nicola Rennie](#), etc.
- Check [Tidytuesday](#), a weekly community of practice.

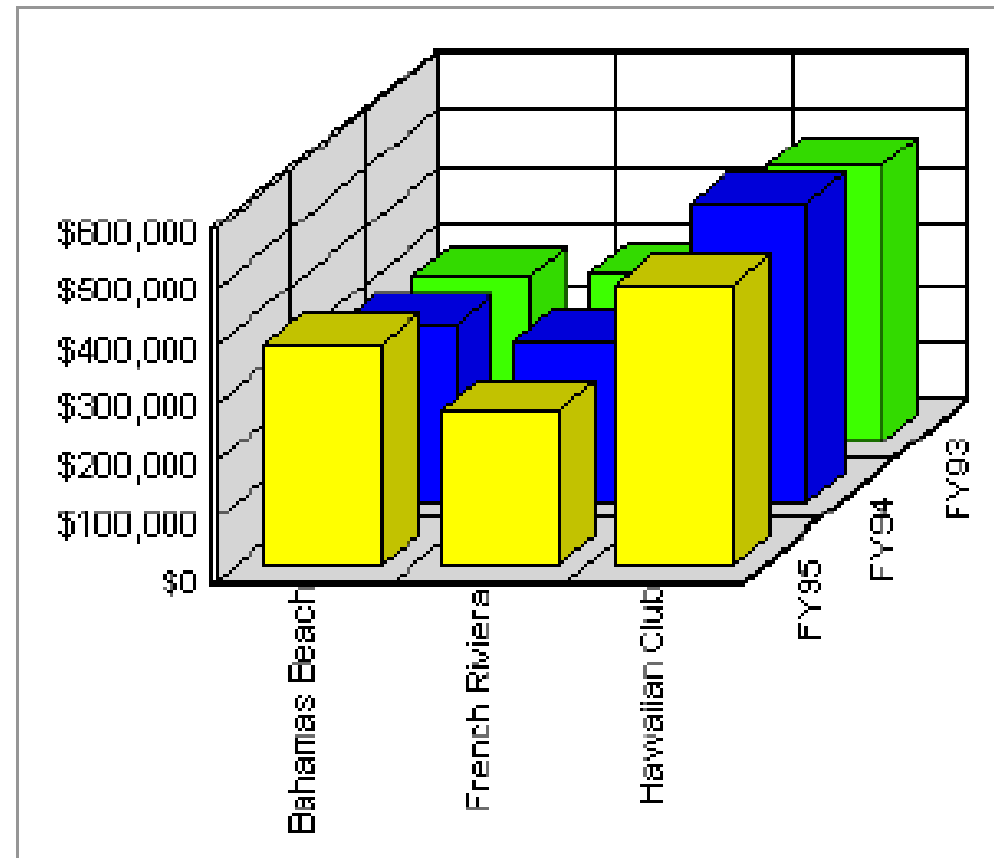
How I see the world after ggplot2



A Design problem (self-study)

A Design Problem

What is wrong with the graph below? Create your own version of the graph. If we have time, we will solve this example in class. If not, try to solve it on your own before you check the detailed solution below.



A Design Problem

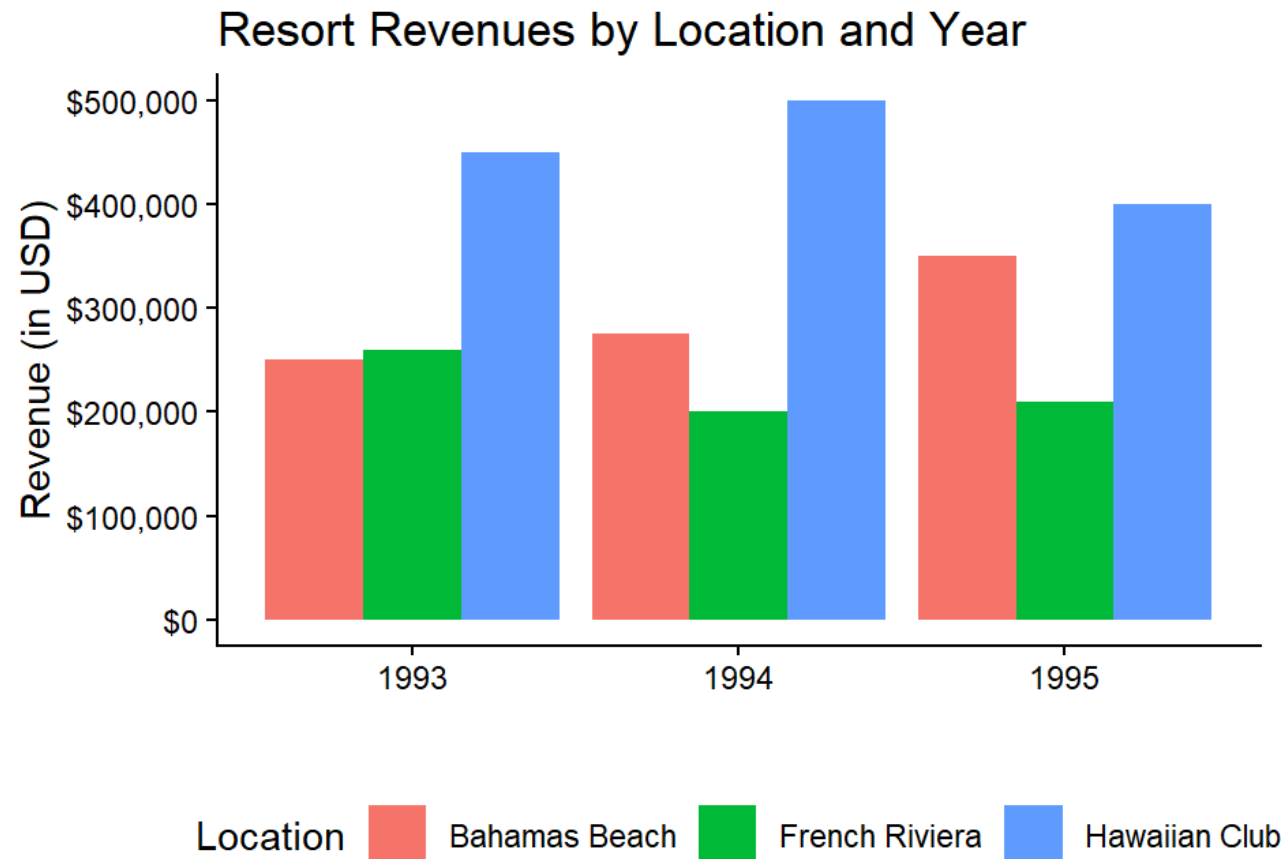
Use the following data to create your own version of the graph:

```
dataChallenge <- data.frame(  
  Location = rep(c("Bahamas Beach", "French Riviera", "Hawaiian Club"), each = 3),  
  Fiscal_Year = rep(c("FY93", "FY94", "FY95"), times = 3),  
  Revenue = c(  
    250000, 275000, 350000, # Bahamas Beach (FY93, FY94, FY95)  
    260000, 200000, 210000, # French Riviera (FY93, FY94, FY95)  
    450000, 500000, 400000 # Hawaian Club (FY93, FY94, FY95)  
  )  
)
```


The problem with this graph

- The 3-D bars are impossible to read.
- The heavy grid lines offer nothing but distraction.
- The vertically-oriented labels (i.e., the resort names and years) are difficult to read.
- The years run from back to front, which is counter-intuitive.

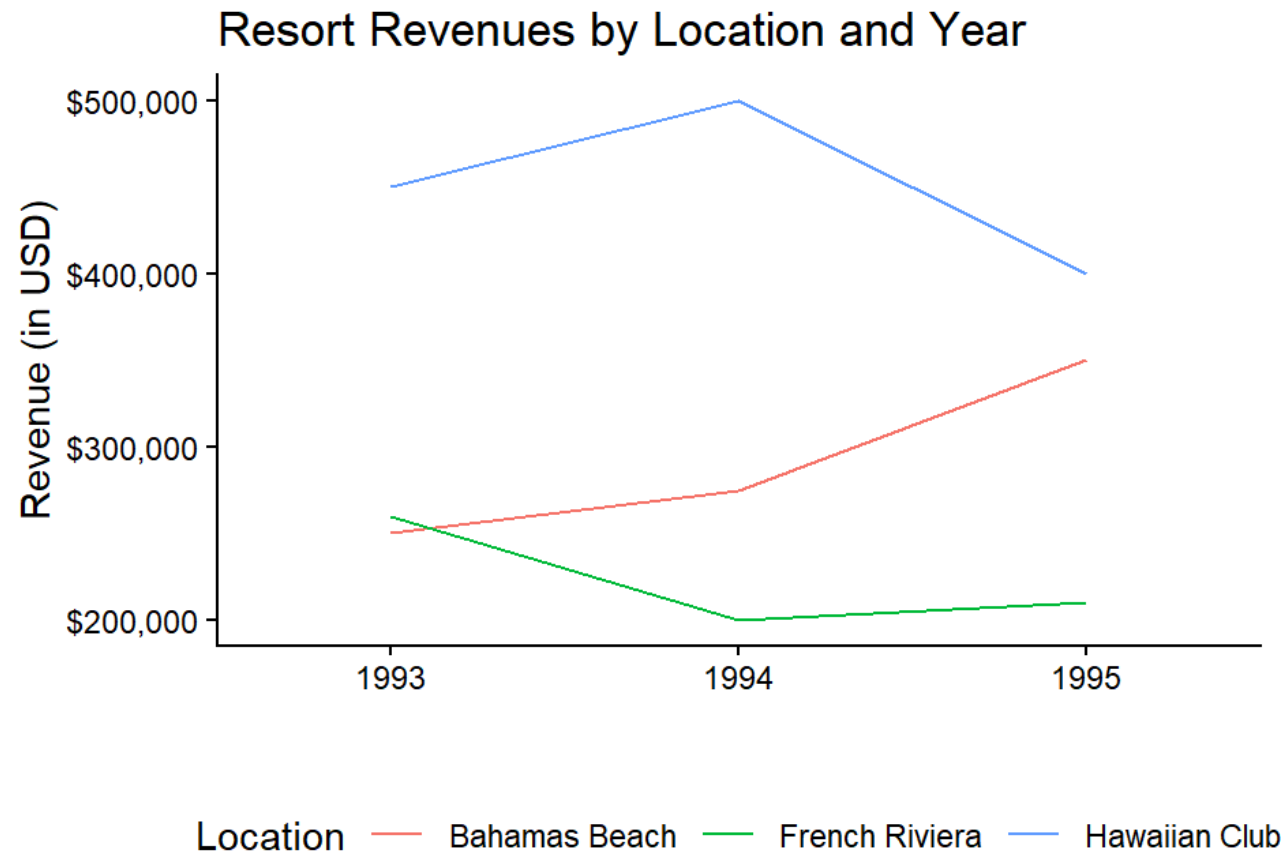
A first solution: comparative performance



- The three resorts have been arranged in order of rank, based on revenue, to highlight their comparative performance.
- The years have been arranged from left to right, which is intuitive.
- The legend has been placed below the bars.

► Show code

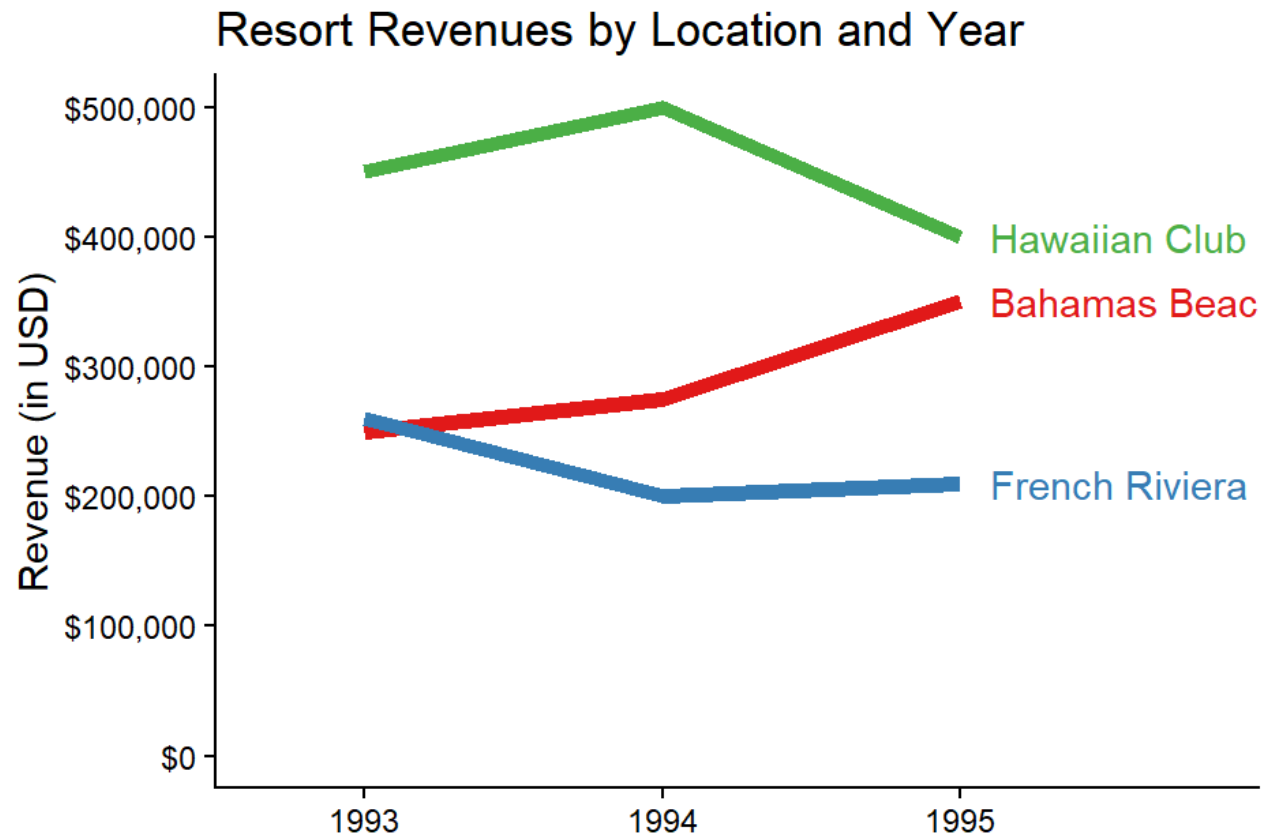
A second solution: change of revenue over time



- This design makes it easy to see how revenue has changed from year to year at each of these resorts.
- However, the magnitudes are difficult to read because the y-axis does not start at 0!
- The eye is still going back and forth between the lines and the legend.

► Show code

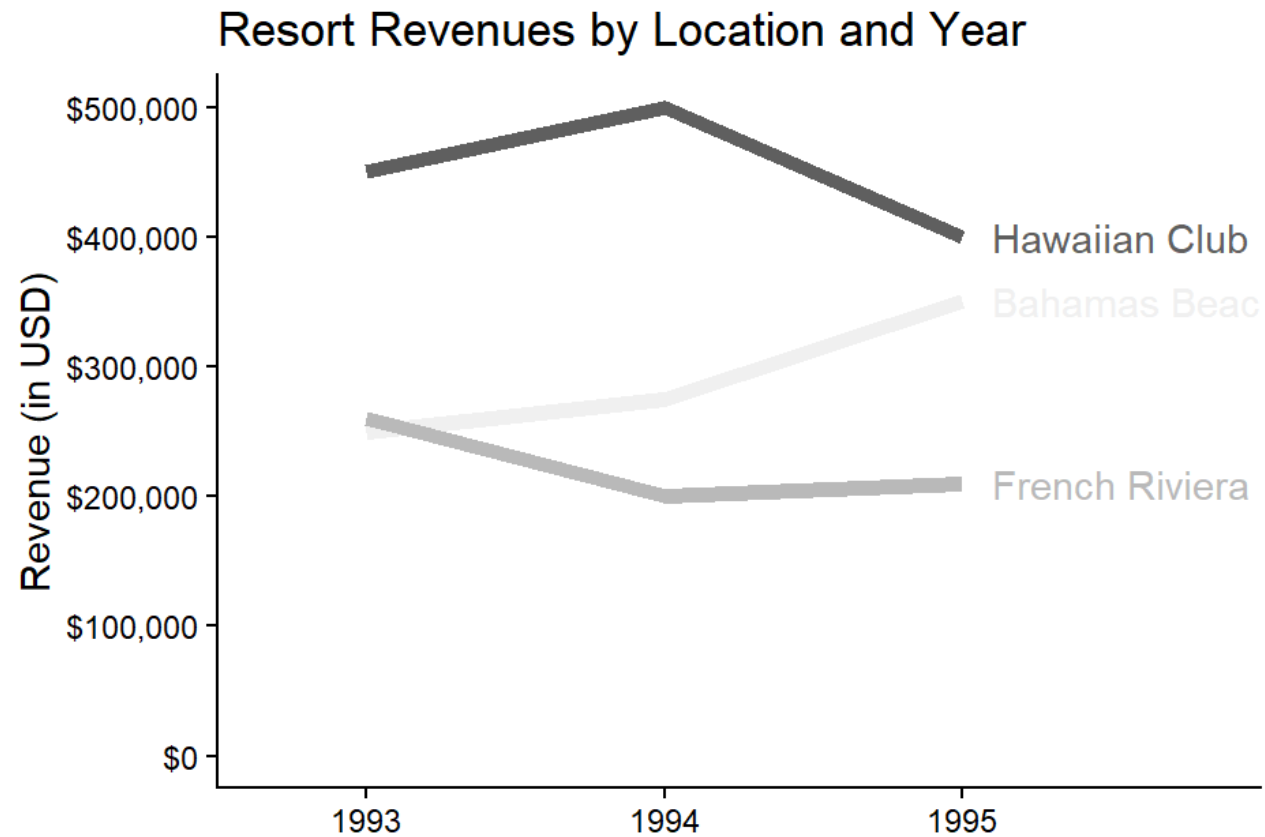
A second solution: change of revenue over time



- This design makes it easy to see how revenue has changed from year to year at each of these resorts.

► Show code

A second solution: change of revenue over time



- Different color palette.

► Show code

Conclusion

Data visualization is an art of story-telling, deception, and scientific exactitude 🤪.